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Consequences of adopting an expanded auditor's report in the United Kingdom

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Abstract The United Kingdom has recently required an expanded auditor's report for large public companies. We investigate whether this requirement is associated with an increase in the decision usefulness of the auditor's report and whether it has indirect consequences on audit fees and quality. Our analyses cover four years surrounding the changes, including companies that transitioned to the new regime and companies that continued issuing the previous report's format. We do not find evidence that the regulatory change significantly affected investors' reaction to the release of auditors' reports, audit fees, or audit quality. Furthermore, we do not find that variation in the expanded reports' content has affected these outcomes. Although companies with long reports pay comparatively higher fees, the mere increase in disclosure does not affect audit fees or quality. Collectively, our evidence is consistent with the expanded auditor's report providing little incremental information to investors.

Keywords Auditor report · Audit opinion · Market reaction · Audit quality · Audit fees

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1 Introduction

The auditor's report is the primary communication between a company's auditor and its shareholders. In most countries and for the vast majority of companies, the auditor's report uses standardized language. Notably, the report provides little information about the company beyond the auditor's opinion, which consists of a pass-or-fail statement regarding the financial statements' compliance with the applicable financial reporting framework.[1] The standard unqualified auditor's report does not communicate potentially useful information about the audit, such as complex issues that require significant judgment and the ways in which the audit addresses them.[2]

Audit regulators and standard setters worldwide are taking steps to expand the content of the auditor's report. In June 2013, the Financial Reporting Council (FRC) in the United Kingdom issued International Standard on Auditing (ISA) 700 (UK and Ireland, revised June 2013), "The Independent Auditor's Report on Financial Statements" (FRC 2013a). Following the FRC's rule changes, the auditor's report must describe the risks of material misstatement that had the greatest effect on the audit, the application of materiality, and the scope of the audit.[3] In our study, we investigate whether the primary objective of the FRC's rule changes has been achieved and whether the rule changes had indirect consequences.

The primary objective of the expanded auditor's report is to improve the informational value of an audit for users of financial statements by promoting greater transparency about the judgments made by management and auditors while preparing and auditing financial statements (FRC 2012c). This objective motivates our first research question: *is the adoption of the expanded auditor's report in the United Kingdom associated with a change in the decision usefulness of the annual report's disclosures for equity market participants?* Previous studies demonstrate that nonstandard auditor reports, such as going-concern opinions, have information content when they are not expected (Frost 1997; Chen et al. 2000; Taffler et al. 2004; Citron et al. 2008; Ghicas et al. 2008; Menon and Williams 2010).

Although generally supportive of the expanded auditor's report, stakeholders were concerned about indirect consequences of implementing the new rule. One of those concerns was the potential increase in prices for audits. The debate between FRC, audit firms, and investors on whether the costs of audits would increase motivates our second research question: *is the adoption of the expanded auditor's report in the United Kingdom associated with a change in audit fees?* The magnitude of the audit fee increases was

[1] When the auditor concludes that the financial statements comply with the applicable financial reporting framework, the auditor's opinion is called unqualified (FRC 2013a; PCAOB 2017) or unmodified (IAASB 2015a). For simplicity, we use the term "unqualified."

[2] Although the auditor is permitted and, in some circumstances required, to modify the wording of the auditor's report while still expressing an unqualified opinion, modifications are relatively narrow in nature. For instance, Czerney et al. (2014, p. 2116) note: "Additional explanatory language typically refers to the adoption of new accounting standards, changes in accounting methods, restatement of prior financial statements, reliance on other auditors to complete the audit engagement, or any other matters the auditor wishes to emphasize." Moreover, Czerney et al. (2014) demonstrate that the most common type of explanatory language mentions the adoption of new accounting standards and changes in accounting methods. These modifications are qualitatively different from the new expanded report disclosures.

[3] Academics (Bedard et al. 2016) and regulators (FRC 2015; PCAOB 2017) refer to the report including additional client-specific information as the extended or expanded auditor's report. For simplicity, throughout our study we use the term "expanded auditor's report."

expected to be small. Before the rule changes, the auditor was already required by ISA (UK and Ireland) 260 (Revised October 2012) "Communication With Those Charged With Governance" to discuss with the audit committee the scope, materiality, and risks of the audit (FRC 2012d). However, an increase in both auditors' effort and litigation risk (linked to expanded disclosures) may arguably have caused an increase in audit fees (Hay et al. 2006; DeFond and Zhang 2014). Moreover, if the new disclosures increased auditors' perceptions of litigation risk and motivated them to perform additional tests and documentation tasks, audit fees may have increased accordingly.

Another concern was the potential improvement in investors' understanding and scrutiny of the auditor's work. This possibility motivates our third research question: *does the adoption of the expanded auditor's report in the United Kingdom have an indirect effect on audit quality?* On one hand, the FRC did not expect the rule changes to affect the underlying work undertaken by auditors but only its transparency (FRC 2012c). Moreover, the benefits of increased oversight may not be immediate or directly linked to an increase in auditors' effort. On the other, the literature on audit quality has generally demonstrated that reputation, litigation, and regulatory concerns shape the incentives that drive audit quality (DeFond and Zhang 2014). The expanded report may influence audit quality by placing companies and their auditors under more scrutiny, especially when it comes to risk disclosures (Christensen et al. 2014).

In answering our three research questions, we implement a difference-in-differences research design that compares two groups of companies with different auditor's report requirements. The expanded report requirements are mandatory only for companies with a premium listing of equity shares on the London Stock Exchange (LSE) Main Market, with fiscal years beginning on or after October 1, 2012 (i.e., fiscal year-ends on or after September 2013). First, we examine LSE premium companies that *were mandated to adopt* the expanded auditor's report and collect data for the four-year period from September 2011 to September 2015, which includes the two years before and after the adoption of the new rule in September 2013. Next, we also use a control group consisting of companies listed in the LSE Alternative Investment Market (AIM), which *were not mandated to adopt* the expanded auditor's report.

Our two main proxies for investors' reaction are cumulative absolute abnormal returns and abnormal trading volume around the date of the public distribution of the annual report, which includes the auditor's report. Our measure of audit fees is the logarithm of total fees paid to the auditor for audit services. Our main proxy for audit quality is the absolute value of discretionary accruals. Overall, we find no evidence that the expanded auditor's report in the United Kingdom is associated with significant changes in these proxies for investors' reaction, audit fees, and audit quality. We further investigate whether variation in (a) the length of the auditor's report, (b) the total number of risks, (c) the number of unique risks (i.e., those not discussed in the audit committee's report), and (d) the materiality threshold had incremental effects on the change (before and after the adoption of the new rules) in investors' reaction, audit fees, and audit quality.[4] We find little evidence that variation in the

[4] Under the new rule, the length of the auditor's report for premium companies has tripled, increasing from approximately one to three pages. The auditor describes typically a total of four risks, including one unique risk not mentioned in the audit committee report. The average materiality is approximately 0.5% of total assets.

report's content incrementally influenced the effect of the expanded auditor's report on our variables of interest.

Our evidence is consistent with the UK expanded report not providing incremental information to investors in the relatively short run. Besides any research design limitations, a lack of a strong market reaction may be driven by these conditions: (a) an auditor's disclosures may be preempted by other information; (b) the market may believe that an auditor adequately addresses the disclosed risks, previously known to the audit committee; or (c) the market may not understand the implications of the auditor's disclosure. We provide some evidence of (a) by showing substantial overlap between the risks listed in the auditor's and the audit committee's reports. We also provide some evidence of (b) by showing that, before and after the new rule, companies with a relatively high number of risks (known ex post) and long auditor reports pay comparatively higher audit fees, after controlling for such factors as size, profitability, and complexity. This finding is consistent with a positive association between audit risks and effort, at least to the extent that audit fees capture auditor effort. Nevertheless, simply *disclosing* the risks does not itself cause a *change* in audit fees or audit quality. Regarding (c), it is only possible if there are significant market inefficiencies, contradicting the evidence presented in studies demonstrating that investors understand and react to new information in auditors' reports (Menon and Williams 2010).

The United Kingdom is a large and developed market, and the FRC's regulatory actions constitute a unique quasi-experimental setting in which we can investigate a substantial change to the auditor's report. Thus our findings may serve to inform ongoing regulatory actions aimed at enhancing the usefulness of the auditor's report.[5] Concurrently with the FRC's rule change, the International Auditing and Assurance Standard Board (IAASB) across countries and Public Company Accounting Oversight Board (PCAOB) in the United States proposed and subsequently adopted new auditor reporting standards (IAASB 2013; IAASB 2015a, 2015b; PCAOB 2013, 2016, 2017).

Our study also contributes to the literature on the auditor's report model in a number of ways (Bedard et al. 2016; Mock et al. 2013). Most of the research on the expanded auditor's report that may directly inform standard setting is experimental (Christensen et al. 2014; Gimbar et al. 2016; Brasel et al. 2016).[6] The vast majority of the related archival research on the value of the auditor's report, however, examines relatively narrow departures from the standard unqualified report (Menon and Williams 2010; Czerney et al. 2017).

Finally, our study helps in the effort to resolve conflicting findings in recent papers examining the consequences of the United Kingdom's expanded auditor's report (i.e., Lennox et al. 2017; Reid et al. 2017; Reid et al. 2015). First, focusing on investors' reaction to the risk disclosures, Lennox et al. (2017) argue that these disclosures lack incremental information. Reid et al. (2015), on the other hand, report that abnormal

[5] Reviews of the first two years of reports conducted by the FRC involving discussions with investors and auditors suggest that there is room for improvement, for example, by increasing the granularity of risk reporting and providing more explanations regarding the materiality, scope, and risk disclosures (FRC 2015, 2016). A potential threat to the relevance of risk disclosures is the inclusion of too many and boilerplate risks that act as an auditor's disclaimer of responsibility (Brasel et al. 2016; Kachelmeier et al. 2017).

[6] These experimental studies investigate the effect of critical audit matters following the PCAOB's rules (CAMs, which are similar to risks of material misstatement in the UK context) on investors' decisions and jurors' perceptions of the auditor's responsibility and legal liability in case of a misstatement. The external validity and the generalizability of these studies to the United Kingdom is an open question.

volume on the annual report's public release date increases in the first year following the new rules. In contrast, our study finds no evidence that the expanded auditor's report causes an incremental short-term market reaction on the public release date. We use multiple proxies for investors' reaction, examine data from four years surrounding September 2013, and consider alternative control groups. In this setting, it is crucial to control for market-wide trends and provide triangulated evidence from multiple proxies for investors' reaction. During 2013, there was an increase in stock prices and a decrease in trading volume. We also find little evidence that investors' reaction is contingent on the report's content. Our conclusions therefore align with the findings of Lennox et al. (2017). Second, in terms of changes in audit fees, our study is consistent with the work of Lennox et al. (2017) and Reid et al. (2017), indicating that audit fees do not increase significantly. Third, regarding changes in audit quality, despite our best efforts to reconcile our findings with those of Reid et al. (2017), our conclusions still diverge from their conclusion that the new rule improves audit quality.[7]

2 Background and research questions

2.1 Move to a mandated expanded auditor's report in the UK in September 2013

The FRC has required auditors to include additional company-specific information in their reports by issuing ISA (UK and Ireland) 700 (Revised June 2013) "The Independent Auditor's Report on Financial Statements" (FRC 2013a). Specifically, paragraph 19A of ISA 700 requires auditors to (1) describe the risks of material misstatements that had the greatest effect on the audit; (2) disclose how the auditor applied the concept of materiality, including a materiality threshold for the statements as a whole; and (3) explain the scope of the audit, including how it relates to the risks of material misstatements and materiality. This requirement is mandatory for companies with a premium listing of equity shares on the London Stock Exchange (LSE) Main Market for fiscal years beginning on or after October 1, 2012 (i.e., fiscal year-ends on or after September 2013).[8] The new FRC standard complements changes made to the UK Corporate Governance Code in October 2012 (FRC 2012a, 2012b, 2014), which

[7] In Section 5, we explain in detail how our analyses and findings complement those of Lennox et al. (2017), Reid et al. (2017), and Reid et al. (2015). Moreover, we conduct several additional analyses aimed at mitigating concerns that the differences between our findings and those reported by Reid et al. (2015, 2017) are the result of sample selection or research design choices. We find no compelling evidence that our conclusions are significantly influenced by differences between our study and those by Reid et al. (2015, 2017), except for the results on discretionary accruals (audit quality), which are sensitive to the calculation of this proxy and sample years. There is also another related study by Smith (2017), but its main focus is on machine-based textual analysis of the auditor's report. Using audit report readability and tone as proxies for communication value, this study finds that, following the passage of the new United Kingdom rules, audit reports have become easier to read and better reflect the risk-related nature of financial statement audits.

[8] The Main Market is the LSE's principal market for listed companies from the United Kingdom and overseas. Premium listing is only open to equity shares issued by trading companies and closed- and open-ended investment entities. Premium companies must comply with the United Kingdom's highest standards of regulation and corporate governance. In the United Kingdom, the new rules apply to premium companies with listing of equity shares, regardless of whether they are incorporated there or elsewhere. In Ireland, this includes Irish incorporated companies with a primary or secondary listing of equity shares on the Irish Stock Exchange. In our analyses, we focus on companies incorporated in Great Britain.

requires additional disclosures for audit committees. The purpose of the FRC's contemporaneous changes to the UK Corporate Governance Code and to ISA (UK and Ireland) 700 was to reinforce the effectiveness of the stewardship model (FRC 2011).[9]

Concurrently with the FRC's regulatory actions, the IAASB and PCAOB issued proposals for new auditor reporting standards (IAASB 2013; PCAOB 2013, PCAOB 2016). An important new requirement in both the IAASB and PCAOB standards is to communicate information about key audit matters (KAMs in the IAASB proposal) or critical audit matters (CAMs in the PCAOB proposal). Similar to the UK requirements regarding the assessment and disclosure of risks of material misstatement identified by the auditor, the purpose of both KAMs and CAMs is to provide additional information about the auditor's work.

The IAASB issued new auditor reporting standards in January 2015, including ISA 700 (Revised) "Forming an Opinion and Reporting on Financial Statements" and ISA 701 "Communicating Key Audit Matters in the Independent Auditor's Report" (IAASB 2015a, 2015b, 2015c). For audits of financial statements for periods ending on or after December 15, 2016, auditors of listed entities are required to communicate KAMs in the auditor's report. The PCAOB issued Auditing Standard (AS) 3101 "The Auditor's Report on An Audit of Financial Statements When the Auditor Expresses an Unqualified Opinion" in June 2017, significantly changing the form and content of the report (PCAOB 2017).[10] In addition to requiring the auditor to communicate CAMs, the standard enhances certain language in the report, adds statements about auditor independence and auditor tenure, requires section titles, and places the opinion in the first section of the report.[11] The new auditor report model is effective for audits of fiscal years ending on or after December 15, 2017, except for the CAM requirement, which is effective on June 30, 2019 for large accelerated filers and December 15, 2020 for all other audits. These concerted actions are part of an ongoing regulatory agenda that may result in further changes to the auditor's report.

2.2 Changes in the decision usefulness of the annual report's disclosures

The objectives of the FRC's regulatory actions on the expanded auditor's report were stated during preliminary discussions in 2011, in Consultation Papers in 2012 and 2013, and in post-implementation reviews in 2015 and 2016. The FRC intended to take actions guided by its "stewardship model" for the auditor; specifically, it "believes that auditors can and should provide increased insight into the audit process so as to reassure users of financial statements that all material matters have been properly disclosed" (FRC 2011). The primary objective of the regulatory change was "to

[9] These disclosures include (1) a description of the significant accounting issues that the audit committee (AC) considered in relation to the financial statements and how these issues were addressed, (2) an explanation of how the AC assessed the effectiveness of the external audit and the approach to the appointment of the auditor, (3) information about auditor tenure and last tender, and (4) how the AC ensures the objectivity and independence of auditors who provide non-audit services (paragraph C.3.8 of the UK Corporate Governance Code FRC 2012).

[10] The US Securities and Exchange Commission approved the standard on October 23, 2017 (SEC 2017).

[11] We note that, although generally similar, the FRC, IAASB, and PCAOB requirements regarding significant risks of material misstatement (KAMs and CAMs) do not have identical definitions. Please refer to the online Appendix D for a summary of the most recent regulatory developments, including details about the definitions of KAMs and CAMs.

enhance the relevance and value of the audit for users and the public by stimulating greater transparency about the judgments made by management and auditors in the course of preparing and auditing financial information" (FRC 2012c). According to the FRC, investors expected that information about the risks, materiality, and scope of the audit would increase transparency with regard to the auditor's application of auditing standards to each client (FRC 2013c). The FRC's primary objectives relate to our first research question: *is the adoption of the expanded auditor's report in the United Kingdom associated with a change in the decision usefulness of the annual report's disclosures for equity market participants?*

Research has examined investors' reaction to additional information in the auditor's report (i.e., departure from a standard unqualified report). US studies demonstrate that nonstandard reports, such as going-concern opinions, have information content when they are not expected (Menon and Williams 2010). By contrast, Czerney et al. (2017) found little evidence to indicate that investors respond to the auditor's explanatory language on the report's release date and attributed this lack of response to investors' incomplete reactions to explanatory language and to previous disclosure of information. Several studies conducted outside the United States have also examined investors' response to a nonstandard auditor's report. These studies show that the market perceives and responds (negatively) to new information contained in modified auditor reports. This association is found in China (Chen et al. 2000), the United Kingdom (Citron et al. 2008; Frost 1997; Taffler et al. 2004) and, for initial public offerings, in Greece (Ghicas et al. 2008).[12] However, we note that going-concern opinions and additional explanatory language constitute rare events and relatively narrow departures from the standard auditor's report. The FRC's regulatory actions provide a unique quasi-experimental setting in which we can investigate a substantial change to the auditor's report, including information on the auditor's judgments about client risk and materiality. The findings of archival studies on the value of the auditor's report, although suggestive, provide only a weak expectation that the expanded disclosures in the United Kingdom will cause a market reaction.[13]

Finally, the expanded auditor's report must convey incremental information to move stock prices or trading volume (typical indicators of decision usefulness in capital markers studies). The expanded report may directly affect the expectations about a firm's future cash flows and discount rate (e.g., by disclosing unexpected risks) or indirectly update beliefs about the firm's financial reporting quality. Specific to the second alternative, while some studies suggest that information quality is priced into a firm's cost of capital through reduced information asymmetry (Easley and O'Hara

[12] On the other hand, an Australian study (Ogneva and Subramanyam 2007) failed to find an association between going-concern opinions and investors' reaction, potentially reflecting the fact that the incidence of going-concern opinions in Australia is significantly higher than elsewhere and that the market there could anticipate the opinions (Simnett et al. 2016).

[13] There is also a stream of recent experimental research that examines the consequences of the expanded auditor's report, focusing particularly on the effect of critical audit matters (CAMs) on investors' decisions and on jurors' perceptions of the auditor's responsibility and legal liability in case of a misstatement. These studies generally support the notion that the expanded report provides useful information to sophisticated investors and lawyers (Christensen et al. 2014) and affects jurors' perceptions of auditor liability following a misstatement (Gimbar et al. 2016; Brasel et al. 2016; Kachelmeier et al. 2017). For additional discussion of the archival and experimental literature on the auditor's reporting model, please see the reviews by Bedard et al. (2016) and Mock et al. (2013).

2004; Francis et al. 2005), others argue that information risk is diversifiable and not priced (Hughes et al. 2007; Lambert et al. 2007; Core et al. 2008; Duarte and Young 2009). It is unclear whether the expanded auditor's report will affect stock returns or trading volume if it only relates to financial reporting quality. Overall, the impact of the UK expanded auditor's report for investors is an open empirical question.

2.3 Changes in audit fees

Various stakeholders expressed concerns about indirect consequences of the new rules. Ten of the 31 comment letters submitted by audit firms and investors to the FRC in 2013 suggested that the costs of audits would increase. Specifically, more senior-auditor hours appeared to be necessary to review and draft the new auditor's report (e.g., BDO, Grant Thornton, and Deloitte). Other commenters pointed to the potential for an increase in proprietary costs (e.g., Chartered Accountants Ireland and Reeves & Co LLP). Ten of the letters also raised the issue that risks to auditors' reputations could have an effect on audit fees (e.g., Baker Tilly, BDO, and Kingston Smith). Finally, Bill Hicks (an investor) worried that "there would be higher costs, because publication of existing information would require more review by the auditor and more dialogue between auditors and managers, thus reducing the efficiency of the audit" (FRC 2013d). Many investors, however, did not expect that incremental costs would be significant, because any increase in costs would largely be related to the public disclosure of matters likely to have already been fully addressed during the audit and reported to the audit committee. The requirement to discuss significant risks with the audit committee was mandated by ISA (UK and Ireland) 260 (Revised October 2012) (FRC 2012d). The FRC stated specifically that "as the auditor was already required to disclose the information to be included in the audit report to the audit committee the work effort in preparing it and familiarizing the audit committee with it should be minimal" (FRC 2013b). These mixed views motivate our second research question: *is the adoption of the expanded auditor's report in the United Kingdom associated with a change in audit fees?*

The literature has documented a link between audit fees and client size, risk, complexity, and auditors' litigation risk (Badertscher et al. 2014; Carcello and Li 2013; Hay et al. 2006; Seetharaman et al. 2002; Bell et al. 2000). An increase in both auditor's effort and litigation risk linked to expanded disclosures may drive an increase in audit fees. It has certainly been demonstrated that companies with going-concern opinions pay higher audit fees than companies with clean opinions (Hay et al. 2006). Moreover, if the new disclosures increase auditors' perceptions of litigation risk and motivate them to perform additional tests and documentation tasks, audit fees may increase accordingly. Evidence presented in a recent experimental study by Gimbar et al. (2016) of the joint effects of risk disclosures in an auditor's report and the precision of accounting standards suggests that the disclosures increase jurors' perceptions of auditors' liability, and that this increased liability may eventually result in increased audit fees. However, it is not straightforward to generalize these results and conclude that litigation risk will increase under the United Kingdom's new rule.

2.4 Changes in audit quality

There were expectations that the expanded auditor's report would improve investors' understanding of the audit process and possibly increase their oversight of the auditor's work. In our review of the 31 comment letters submitted, 18 commenters mentioned

improving users' ability to assess audit quality. Five others, however, expressed the opinion that the new rule did not increase users' ability to assess audit quality. The FRC emphasizes that the new requirements attempt to make the auditor's work more transparent but do not influence the underlying work undertaken by auditors under existing auditing standards (FRC 2012c). Moreover, the benefits of increased oversight may not be immediately apparent or directly linked to audit hours. For instance, investor Bill Hicks argued that companies may use benchmarking to influence materiality, that the "expectation gap" may increase, and that there could be increased costs that audit firms might not be able to pass on to the client, factors that may negatively impact quality (FRC 2013d).[14] Finally, the expanded auditor's report could attract additional FRC oversight. The FRC first-year review highlights the importance of ensuring that auditors' reports accurately reflect the work performed, identifying inaccuracies in auditors' descriptions of the nature or extent of the audit procedures performed (FRC 2015). These mixed expectations motivate our third research question: *does the adoption of the expanded auditor's report in the United Kingdom have an indirect effect on audit quality?*

The expected link between the auditor's report content and audit quality is somewhat unclear. Studies have used modified opinions as an indicator of audit quality. Thus, for example, after controlling for the client's financial condition and other factors, a high incidence of going-concern opinions among Big N auditors is interpreted as evidence of high audit quality (DeFond and Zhang 2014). The association between financial reporting quality and modifications of auditors' reports is not, however, straightforward. Although Bartov et al. (2000) found a link between earnings management (i.e., discretionary accruals) and modified audit opinions, Butler et al. (2004) were unable to confirm their results. A further perspective is that of market oversight. The literature on audit quality has generally demonstrated that reputation, litigation, and regulatory concerns shape the incentives that drive audit quality (DeFond and Zhang 2014). The expanded report may, then, influence audit quality by placing companies and their auditors under relatively more careful scrutiny, especially when it comes to risk disclosures (Christensen et al. 2014). However, it is unclear whether there is a direct correspondence between expanded disclosures and audit quality. Conceptually, two conditions have to be met to motivate a change in audit quality under the new rule. First, the cost of increasing audit quality is lower than the expected marginal cost of litigation or other penalties (given an ex ante level of audit quality). Second, the expanded auditor's report will reveal suboptimal audit quality previously unknown. Whether the expanded disclosures facilitate users' scrutiny and indirectly affect audit quality is an empirical question.

3 Research design

3.1 Difference-in-differences and pre-post adoption models

We take advantage of the cut-off date for the adoption of the expanded auditor's report requirements, applicable to the LSE Main Market premium listed companies with fiscal

[14] For instance, The National Audit Office considered "the proposed new requirements as illustrated do not provide genuine insight into the audit nor the most useful 'hooks' for investors to use in subsequent discussions with management."

year-ends on or after September 2013. We collect data for premium companies (i.e., rule adopters) for a four-year period from September 2011 to September 2015, spanning two years before and two years after the adoption of the new rules. We note that there were time-related trends in the United Kingdom's stock market during this period. Figures 1 and 2 illustrate the trends in monthly average closing prices and trading volume (in millions of shares) for the UK Financial Times Stock Exchange (FTSE) 100 index from January 2011 to December 2015. The vertical solid line represents the date on which the new auditor's report rule was adopted (i.e., September 2013). The vertical dashed lines mark the four years around the adoption date. Figure 1 shows a noticeable increase in closing prices during the period of our study, specifically in 2013. Fig. 2 shows that the first year of our sample (from September 2011 to September 2012) was characterized by a period of relatively high trading volume.[15] We accordingly implement a difference-in-differences research design to mitigate the effect of time-related trends contemporary with the adoption of the new rule, including economic, political, and other factors. Our control group consists of UK companies listed on the LSE Alternative Investment Market (AIM). The AIM is a segment intended to assist smaller and growing companies in raising external financing. Importantly, AIM companies were not mandated to comply with the new rule, and as part of our data collection, we verified that the AIM companies in our sample used the old auditor's report format during the two years before and two years after September 2013. Figure 3 shows the trend of daily returns in the FTSE100 and Total AIM indexes (ticker symbols ^FTSE and ^FTAI, respectively). These indexes capture common economic shocks, and their daily returns have a positive correlation of 0.68 (statistically significant, $p < 0.01$). As stated earlier, the new rules in general constitute an unexpected and transparent source of variation in the auditor's report content. Moreover, this exogenous shock applies differently to premium and AIM companies, which allows us to estimate the following difference-in-differences model:

$$DEPVAR_{i,t} = \beta_0 + \beta_1 POST_{i,t} + \boldsymbol{\beta_2 ADOPT_{i,t}} + \boldsymbol{\beta_3 POST{*}ADOPT_{i,t}} + \sum \beta_j CONTROLS_{i,t} + IndustryFE + \varepsilon_{i,t}, \quad (1)$$

where, for company i in year t, $DEPVAR_{i,t}$ is one of our proxies for (1) investors' reaction to the public dissemination of the annual report (which includes the auditor's report), (2) audit fees, and (3) audit quality. $\boldsymbol{ADOPT_{i,t}}$ is an indicator variable for our treatment and control firms that is equal to one for premium companies and zero for the AIM companies. $\boldsymbol{POST_{i,t}}$ is a period indicator variable that equals one for fiscal year-ends on or after September 2013 and zero otherwise. Finally, $\boldsymbol{POST{*}ADOPT_{i,t}}$ is an interaction of the treatment and time indicator variables that captures the difference-in-differences effect. We also perform three

[15] The FTSE 100 index is maintained by the FTSE Russell Group This index includes the 100 companies listed on the LSE with the highest market capitalization (representing approximately 80 percent of the total LSE capitalization). This index is a widely used summary indicator for the UK stock market. For additional details, please see http://www.ftserussell.com. In untabulated analyses, we also examined the trends in closing prices and trading volume for the FTSE 250, a similar index including the 101st to the 250th companies listed on the LSE with the highest market capitalization. The trends are qualitatively similar to those for the FTSE 100. In 2013, other European indexes also experienced an upward trend, reporting their best annual gains since 2009. France's CAC-40, Spain's IBEX, and Germany's DAX increased by 18, 20, and 26 percent, respectively. The FTSE 250 closed at 15,935 and gained 28.8 percent over 2013 (Allen 2013).

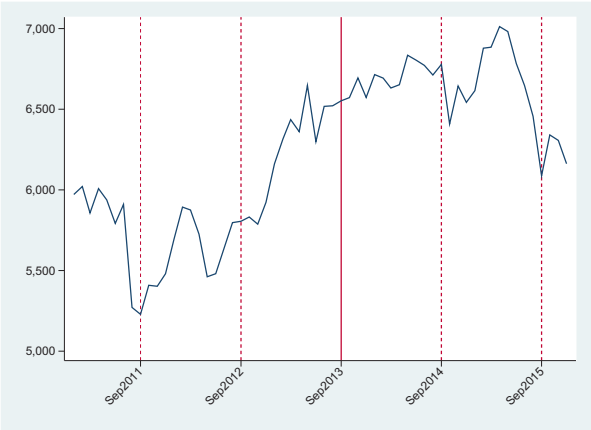

Fig. 1 FTSE100 index closing prices (Monthly Average)

analyses to complement our main model. First, we calculate standard errors using the bootstrap method (with 1000 replications), which has proved helpful in cases where the theoretical distribution of the test statistic is unknown and the sample sizes are small. Second, we include company fixed effects in Eq. 1 and exclude the $ADOPT_{i,t}$ intercept, thus mitigating the impact of time-invariant company characteristics and strengthening the difference-in-differences design. (See Angrist and Pischke 2009 for additional details.) Third, we estimate the pre-post adoption effect for the group of premium companies only. The pre-post model provides additional evidence as to whether any effect is driven by variation in the treatment group alone.

3.2 Analyses of investors' reaction to the auditor's report

The auditor's report is included in the release of each company's full annual report. The UK Companies Act mandates that annual reports be made publicly available. Specifically, the UK Financial Conduct Authority's Disclosure and Transparency Rule (DTR) 6.3 requires a public announcement to be issued when the report is submitted to the

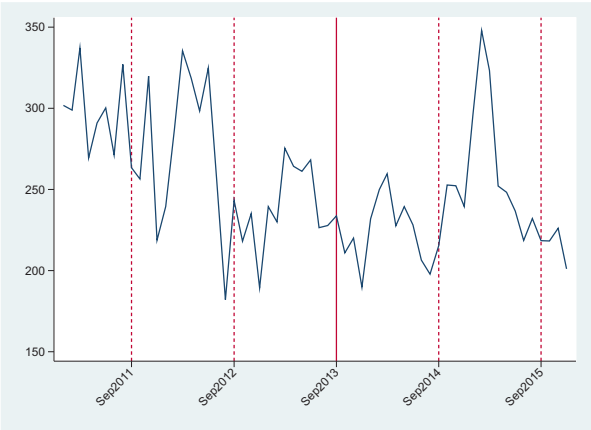

Fig. 2 FTSE100 index trading volume (Monthly Average in Millions)

National Storage Mechanism and is made available for viewing on every company's website. We searched company announcements of annual filings using the Regulatory News Service (RNS) of the LSE as well as the National Storage Mechanism (NSM).[16] The release of the full annual report follows an earlier earnings announcement disclosure. For the premium companies in our sample, the earnings announcement represents a very complete form of disclosure, containing full financial statements and mentioning whether the auditor's opinion was unqualified. Moreover, the listing rules require preliminary announcements to be agreed upon by the auditor prior to publication. Appendix 2 illustrates our method for identifying and calculating daily returns in the period surrounding the public release dates (i.e., filing dates) of annual reports.

Our two main proxies for investors' reaction are (1) the cumulative absolute abnormal returns in the period surrounding the date on which the annual report, containing the auditor's report, is publicly distributed (i.e., the report filing date) and (2) the sum of three-day abnormal trading volume around the report filing date. Absolute abnormal returns capture the average change in investors' belief in response to an announcement event. Abnormal trading volume reflects the change in individual investors' beliefs (Garfinkel and Sokobin 2006). Trading volume, however, arguably captures information content as well as information asymmetry and investor disagreement regarding new information. The literature suggests that, while returns and volume correlate positively to some extent, the two measures can be considered largely independent (Bamber and Cheon 1995). Further, any given information event can result in high abnormal volume and almost no change in returns or vice versa (Kandel and Pearson 1995). Using these measures together increases our chances of detecting and interpreting investors' reaction to the new auditor's report. Particularly, abnormal returns help to discriminate between two primary reasons for abnormal trading volume, information content, or disagreement (Bamber et al. 2011). Furthermore, using both returns and volume is important, given the relatively small size of our samples (Cready and Hurtt 2002).

We calculate (1) absolute abnormal returns, $ABRET_{i,t}$, as the sum of the three-day absolute abnormal returns around the filing date by company calculated each day as the company returns = (Price Close_t – Price Close_{t-1})/Price Close_{t-1} minus same-day returns for the LSE value-weighted portfolio, and (2) abnormal volume, $ABVOL_{i,t}$, as the natural logarithm of the firm's average event-period volume divided by the firm's mean estimation-period volume. Event-period volume is calculated as the daily volume over the three-day event window around the report filing date and is scaled by shares outstanding. The estimation-period volume is measured over the trading period beginning 61 days before the earnings announcement date and ending 40 days later (i.e., 21 days before the

[16] For example, see the announcement by ITE Group plc on December 24, 2013, in RNS that its annual report is available at http://www.londonstockexchange.com/exchange/news/market-news/market-news-detail/11816292. html. The FRC requirements are described in detail at http://fshandbook.info/FS/html/handbook/DTR/6/3. We were unable to identify reliable report filing dates in Worldscope, Bloomberg, or other online open sources. We find that the annual report filing dates in Worldscope and Bloomberg often correspond to the earnings release date, as per the RNS announcements. The LSE RNS website can be accessed searching each company at http:// www.londonstockexchange.com/prices-and-markets/markets/prices.htm. The NSM can be accessed searching each company at http://www.morningstar.co.uk/uk/NSM. An alternative source is the website http://www.investegate.co.uk. For a small number of observations, we are unable to identify the specific annual report release date. In these cases, we use the Annual General Meeting announcement date, if available, or the Annual General Meeting date.

Fig. 3 Trends in Daily Returns of FTSE100 and Total AIM Indexes

earnings announcement date).[17] The list of control variables for these analyses includes $LOGMKT_{i,t}$, the natural logarithm of total market value; $ROA_{i,t}$, the net income before extraordinary items/total assets; $LOSS_{i,t}$, an indicator variable equal to one if ROA is negative and zero otherwise; $MTB_{i,t}$, the equity market value/book value; $LEV_{i,t}$, the long-term debt/total assets; $LAG_{i,t}$, the days between the earnings release and annual report filing date; $SALEVOL_{i,t}$, the standard deviation of sales/total assets over year t to year t-6; $CHNI_{i,t}$ (net income before extraordinary items in year t - net income before extraordinary items in year t-1)/total assets; $BETA_{i,t}$, a slope coefficient obtained by regressing the daily return of a company's stock on daily returns of the value-weighed portfolio over the 220-day period (−250, −21), relative to the filing date; $ABRET_EAR_{i,t}$, the sum of the three-day absolute abnormal returns around the earnings release date by company; and $BIG_{i,t}$, an indicator variable equal to one if the company has a Big 4 auditor and zero otherwise. Our list of control variables follows Carcello and Li (2013, Table 3, Panel A), who examine changes in earnings informativeness after the regulatory changes in the United Kingdom requiring the disclosure of partners' names. We also include annual change in net income and three-day absolute abnormal returns around the earnings release date as controls for earnings news and beta as a control for company risk.

3.3 Analyses of audit fees

Our measure of audit fees is the logarithm of total fees paid to the auditor for audit services in each year, $FEES_{i,t}$. We manually obtained audit fees from the notes to the annual financial statements. We include a set of control variables, commonly used in previous audit fee research, to isolate the effects of client risk, size, and complexity (Hay et al.

[17] We calculate market variables by aggregating companies' individual data. We obtained the constituents of the LSE from the WSCOPEUK list from Worldscope for securities on LSE that are major and primary (EXDSCD = 'LN' MAJOR = 'Y' and ISINID = 'P') and that are neither investment trusts nor closed-end funds (TYPE = 'EQ'). This list includes both Premium and AIM listed companies. Our measure of abnormal volume is consistent with those of DeFond et al. (2007) and Miller (2010).

2006). The list of control variables for these analyses includes $SIZE_{i,t}$, the natural logarithm of total assets; $ROA_{i,t}$; $LOSS_{i,t}$; $MTB_{i,t}$; $LEV_{i,t}$; $CFO_{i,t}$, cash flow from operations/total assets; $SALEVOL_{i,t}$; $BIG_{i,t}$; $INV_{i,t}$, inventory/total assets; $REC_{i,t}$, receivables/total assets; $FORSALE_{i,t}$, foreign sales/total assets; $BUSY_{i,t}$, an indicator variable equal to one if fiscal year-end is between December and March and zero otherwise; $GCO_{i,t}$, an indicator variable equal to one if the auditor issued a going-concern opinion and zero otherwise; $USLIST_{i,t}$, an indicator variable equal to one if the company is cross-listed in the United States and zero otherwise; and $NSEG_{i,t}$, the natural logarithm of the number of business segments. Again, our list of control variables follows that of Carcello and Li (2013, Table 3, Panel B) in examining changes in fees after the regulatory changes in the United Kingdom that required the disclosure of partner names. We also include two additional control variables that commonly appear in audit fee models, namely an indicator for going-concern opinions and number of segments (e.g., Boone et al. 2017). Finally, we include an indicator for United States cross-listed status, based on the findings of Seetharaman et al. (2002), who document an audit fee premium for these companies.

3.4 Analyses of audit quality

Our proxy for audit quality in our main analyses is the absolute value of discretionary accruals, $DACCR_{i,t}$, which is estimated using the Jones model, including ROA and scaling all variables by average assets.[18] This measure captures audit quality with respect to financial reporting, with higher audit quality being defined as greater assurance that the financial statements faithfully reflect a firm's underlying economics given the financial reporting system and its innate characteristics (DeFond and Zhang 2014). This measure has been used extensively in research on audit quality examining companies from the United States and other countries (Lawrence et al. 2011; Carcello and Li 2013; Boone et al. 2017; Fung et al. 2017; Lamoreaux 2016). Our list of control variables follows that of Carcello and Li (2013, Table 3, Panel A) and includes $SIZE_{i,t}$, $ROA_{i,t}$, $LOSS_{i,t}$, $MTB_{i,t}$, $LEV_{i,t}$, $CFO_{i,t}$, $SALEVOL_{i,t}$, $BIG_{i,t}$, and $LAGACCR_{i,t}$ (the total accruals in year t-1).[19]

[18] Discretionary accruals are estimated using an annual cross-sectional model for each SIC two-digit industry (Kothari et al. 2005) for all the companies on the London Stock Exchange. To obtain information about these companies, we use the WSCOPEUK list from Worldscope for securities on LSE that are major and primary (EXDSCD = 'LN' MAJOR = 'Y' and ISINID = 'P') and that are not investment trusts or closed-end funds (TYPE = 'EQ'). Discretionary accruals are the residual of the following model: $TACCR_{i,t} = \alpha + \beta_1 ROA_{i,t} + \beta_2 \Delta Sales_{i,t} + \beta_3 PPE_{i,t} + \varepsilon_{i,t}$, where, for firm i and year t, $TACCR_{i,t}$ is income before extraordinary items - cash flow from operations)/average total assets; $ROA_{i,t}$ is (net income before extraordinary items)/average total assets; $\Delta Sales_{i,t}$ is (sales in year t - sales in year t-1)/average total assets; and $PPE_{i,t}$ is (gross property, plant, and equipment)/average total assets. The model is estimated by industry (SIC two-digit), excluding industries with less than 20 observations.

[19] In the context of our study, it is difficult to implement analyses of other common audit-quality proxies. Going-concern opinions have a very low incidence; we find only 21 going-concern opinions among 1212 firm-year observations used in our audit fee analyses. Financial restatements are also rare and arguably have limited consequences. Each year, the FRC typically examines 300 annual reports from all listed companies and asks about 15 to 20 companies either to restate their accounts or to change their practices for the following year. However, some observers claim that "such an FRC requirement will not attract a wider audience as it will be mentioned only briefly in a company's report and accounts" (Smith 2014). In our additional analyses section, we discuss results using other dependent variables for audit quality, including meeting or beating the analyst consensus forecast and earnings response coefficients (ERCs). However, we note that, despite our best efforts, all our proxies for audit quality are inherently indirect.

3.5 Cross-sectional analyses using the content of the auditor's report

We further investigate whether variation in (a) the length of the auditor's report, (b) the total number of risks, (c) the number of unique risks (i.e., those not discussed in the audit committee's report), and (d) the materiality threshold had an incremental effect on the change (pre- and post-adoption) in investors' reaction, audit fees, and audit quality. These analyses constitute a deeper examination of the effect of the new rules. The expanded auditor's report could have affected a subset of companies, depending on their reports' content, even when there might not be a clear average pre-post adoption effect. For premium companies only, we estimate the following cross-sectional model (before and after the adoption of the new rules), taking into account variation in the reports' content:

$$DEPVAR_{i,t} = \beta_0 + \beta_1 POST_{i,t} + \beta_2 REPORT_DISCL_{i,t} + \beta_3 POST^{*} REPORT_DISCL_{i,t} + \Sigma\beta_j CONTROLS_{i,t} + IndustryFE + e_{i,t}, \quad (2)$$

where, for company i in year t, $REPORT_DISCL_{i,t}$ is one of four variables capturing cross-sectional variation in the auditor's report length, number and nature of risks, and materiality.[20]

Taking each of these variables in turn, $LENGTH(1/0)_{i,t}$ is an indicator variable that is equal to one if the average number of words in the auditor's report (i.e., its length) in the adoption years t and $t+1$ exceeds the 50th percentile of the distribution for all companies and zero otherwise. After controlling for company characteristics, including size, profitability, and sales volatility, our $LENGTH(1/0)_{i,t}$ indicator variable is intended to capture the incremental effect of long reports on our dependent variables. On the one side, the FRC's intention was for comparatively longer and more detailed reports to provide additional useful information (FRC 2013d). On the other side, Bloomfield (2002) and Li (2008) have argued that the incomplete revelation hypothesis implies that managers can reduce the market response to bad news by making it more costly to analyze. The evidence presented by Li (2008) suggests that managers make bad news costly by writing verbose annual reports that feature jargon and long sentences.

Second, $NRISKS(1/0)_{i,t}$ is an indicator variable that is equal to one if the average number of risks mentioned in the auditor's report in years t and $t+1$ exceeds the 50th percentile of the distribution and equal to zero otherwise. After controlling for company characteristics, including size, profitability, and sales volatility, our $NRISKS(1/0)_{i,t}$ indicator variable is intended to capture the incremental effect of a large number of risks on our dependent variables.[21] On the one side, the auditor's report typically includes risks relating to specific areas, for example, uncertainty regarding revenue recognition, tax accounting, or valuation. Thus the identification of a large number of

[20] We do not explicitly assess the effect of scope disclosures, apart from examining the effects of the total length of the auditor's report, because they are difficult to quantify, in many cases discussing the group structure of the audit without providing additional details.

[21] We consider the risk section language a separate issue. Two similar companies can have standardized or boilerplate language in their risk disclosures, but one company reports four risks and the other reports only one. There is still variation in disclosure; arguably four risks are a more severe threat than one. We believe it is reasonable to expect that the number of risks matters, despite the language used to explain each risk. A concurrent study by Smith (2017) examines variation in readability and tone of the auditor's report language.

risks may indicate pervasive risk (i.e., risk impacting financial reporting in multiple accounts).[22] On the other side, Kachelmeier et al. (2017) find that the disclosure of critical matters might cause users to perceive the matters as more complicated and subjective and make the disclosure a disclaimer reducing auditor responsibility for subsequently discovered misstatements. Thus the auditor might include more risks. Overall, our expectation with respect to $NRISKS(1/0)_{i,t}$ are similar to those for $LENGTH(1/0)_{i,t}$. The discussion of risks is viewed as a central part of the expanded auditor's report.

Third, $NRISKSAUD(1/0)_{i,t}$ is an indicator variable that is equal to one if the average number of unique risks in the auditor's report (again, those not mentioned in the audit committee report) in years t and $t+1$, exceeds the 50th percentile of the distribution and is equal to zero otherwise. There were other changes to corporate reporting in the United Kingdom around the same time, in particular, those to the UK Corporate Governance Code in October 2012 requiring additional disclosures by the audit committee of the board of directors in the corporate governance section of the annual report (FRC 2012b). The audit committee report includes a list and brief discussion of significant accounting risks. We recognize that it is inherently difficult to disentangle the audit committee and auditor reports, between which there is a large degree of overlap. Our expectations for $NRISKSAUD(1/0)_{i,t}$ are similar to those for $NRISKS\ (1/0)_{i,t}$. However, $NRISKSAUD(1/0)_{i,t}$ is intended to capture the incremental impact of unique risks discussed by auditors. We manually coded this variable while reading the reports from each company.

Fourth, $MATPERC\ (1/0)_{i,t}$ is an indicator variable that is equal to one if the average materiality in British pounds, scaled by total assets, in years t and $t+1$ exceeded the 50th percentile of the distribution and equal to zero otherwise. This variable is intended to capture the incremental effect of a large materiality threshold on our dependent variables. As noted by KPMG (2014, p. 16), "a wealth of information about audit materiality is now entering the public domain for the first time. This is a further step forward in the transparency of audit." The KPMG report further explained the impact of materiality on audit work as follows: "risk and materiality are both drivers of the audit work. The volume of work is determined by a combination of the assessment of risk and the materiality. Increasing the risk assessment increases the work done at any given level of materiality." Moreover, investors who support the new auditor's report have argued that audit judgment (e.g., the auditor's assessment of risks and materiality) can increase the understanding of how audit firms apply auditing standards and can also aid investors in evaluating the quality of an audit (FRC 2013d). Some investors, however, have raised the concern that the disclosure of materiality may encourage competitive benchmarking toward the lowest standard (i.e., the highest materiality threshold) and, in this way, undermine audit quality (FRC 2013d).

[22] Although large companies, being in most cases relatively more complex, tend to report more risks than smaller ones, these risks are widely scattered across companies of various sizes. For example, a descriptive report by KPMG (2014, p. 18) notes: "auditors of many FTSE 300+ companies are reporting just as many risks as the auditors of some FTSE 50 companies."

4 Sample selection and results

4.1 Sample selection

Table 1 shows our sample selection. We began by considering all 728 companies incorporated in Great Britain that have ordinary stocks listed in the LSE premium category and traded on the LSE Main Market. We merged this list of companies with the Thomson Reuters databases to identify 338 nonfinancial companies with data in Worldscope and Datastream (i.e., eliminating SIC codes 6000 to 6999). This selection was made because many banking, insurance, trust, and investment companies are traded in the LSE Main Market (KPMG 2014), while our study focuses on nonfinancial companies, as has the literature (e.g., Carcello and Li 2013; Czerney et al. 2017; Seetharaman et al. 2002).[23] We read and manually coded information from annual reports (e.g., auditor's report length, names of audit firm and audit partner, number and description of significant risks of material misstatement in the auditor's report, materiality amount, audit fees, and the number and descriptions of significant accounting issues in the audit committee report). Finally, we complemented our data with financial and market variables from the Thomson Reuters databases (Worldscope, Fundamentals, and Datastream). We also identified the date on which each company's annual report was made available to investors (i.e., the filing date) and the earnings release date in the LSE RNS and NSM databases. Similar to the sample selection for premium companies, we started with 724 AIM companies listed in the LSE and incorporated in Great Britain and then excluded financial companies and those without data in Thomson Reuters. We found 525 companies that met these criteria, hand-collected the names of audit firms, partners, and fees from annual reports, and complemented these data with financial and market variables from the Thomson Reuters databases (Worldscope, Fundamentals, and Datastream). We also again hand-collected the date on which each company's annual report was made available to investors (i.e., the filing date) and the earnings release date in the LSE RNS and NSM databases. We acquired data for all of our variables for the full four-year panel in the period surrounding the regulatory cut-off in September 2013. Table 1 shows the number of observations for each of our difference-in-differences analyses.[24] In our additional cross-sectional analyses, we use premium companies only.[25]

[23] Excluding financial firms is a typical practice in audit fees and audit quality research. Banking and other financial companies (e.g., mutual funds) face high levels of regulatory scrutiny and have additional mandatory disclosures. There are no straightforward reasons to expect that the expanded auditor's report will be incrementally informative for these companies. We reviewed a number of reports for these companies and found that the most common risk mentioned is valuation of assets, focusing on determining the fair value of investments.

[24] The data used in our analyses are provided in online Appendix C.

[25] The companies listed in the LSE are available at http://www.londonstockexchange.com/statistics/companies-and-issuers/companies-and-issuers.htm. We used the list available on April 2015. We identified only three early adopters in our sample. Vodaphone Group included an auditor's report based on the FRC's exposure draft for fiscal year-end March 31, 2013. Sky plc included an auditor's report based on the revised FRC standard issued in June 2013 for fiscal year-end June 30, 2013. Ashmore Group included an auditor's report based on the revised FRC standard issued in June 2013 for fiscal year-end June 30, 2013. Vodaphone Group and Sky plc are included in our sample. For these companies, consistent with the September 2013 cut-off, we considered March 31, 2014, and June 30, 2014, as the first regular year of adoption. We eliminated from our sample a few cases in which companies did not disclose a specific materiality amount, audit fees were not disclosed, or we were unable to extract the auditor's report from the annual report. We used the list of LSE AIM companies available on July 2016.

Table 1 Sample selection

	LSE Main Market with premium listing	LSE AIM	DD Sample
UK companies on the LSE	728	724	
Exclude financial companies	(369)	(99)	
Exclude companies not in Thomson Reuters	(21)	(100)	
Nonfinancial companies in Thomson Reuters	338	525	
UK companies on the LSE with data available for full panel (four years around September 2013)			
Investors' reaction analyses	312	328	
Audit fees analyses	303	360	
Audit quality analyses	218	296	
Firm-year observations:			
Investors' reaction analyses	1248	1312	2560
Audit fees analyses	1212	1440	2652
Audit quality analyses	872	1184	2056

This table shows the sample selection for the analyses of investors' reaction, audit fees, and audit quality. We use company data for four years in the period surrounding the regulatory cut-off in September 2013. In our main difference-in-differences analyses, we use premium companies as the treated group and AIM market companies as a control group. In our additional cross-sectional analyses, we use premium companies only. The data used in our analyses are provided in online Appendix C

4.2 Investors' reaction to the auditor's report

Figure 4 illustrates the year-over-year distribution of cumulative absolute abnormal returns $ABRET_{i,t}$ in the period surrounding the date of the public release of the annual report using a box plot. The graph separates the AIM and premium companies by fiscal year. Year t is the first fiscal year following the adoption of the new rules. Each box represents the 25th to 75th percentile range, and the diamond dot represents the 50th percentile (median). The labels denote the 25th, 50th, and 75th percentile values. The vertical lines (whiskers) represent the distribution of values below the 25th and above the 75th percentiles and limited at the adjacent values as recommended by Tukey (1977), which are equal to the 25th and 75th percentile minus and plus 1.5 times the interquartile range, respectively. Figure 4 demonstrates that, over the years covered in our study, cumulative absolute abnormal returns were relatively small and stable for both AIM and premium companies. Specifically, over the four years covered by our sample, the median $ABRET_{i,t}$ for premium companies was, respectively, 0.014, 0.014, 0.016, and 0.015; the 75th percentile was 0.028, 0.029, 0.029, and 0.028; and the 25th percentile was 0.007, 0.007, 0.008, and 0.007. Fig. 4 also shows that the cumulative absolute abnormal returns of the AIM group had a comparatively larger dispersion (i.e., larger boxes and whiskers). Next, to provide an idea of the distribution of positive and

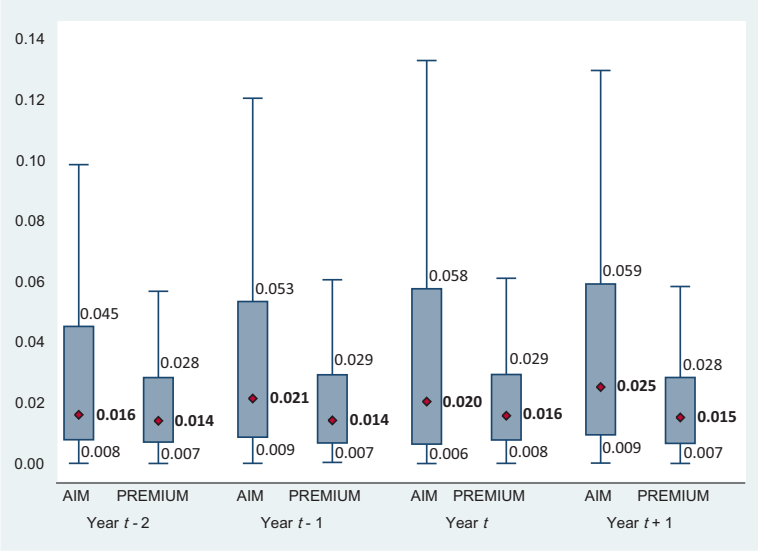

Fig. 4 Absolute abnormal returns (*ABRET*)

negative returns, Fig. 5 shows signed cumulative abnormal returns and also suggests a stable trend for both AIM and premium companies. Thus, for instance, in the case of premium companies, the median signed cumulative abnormal returns were − 0.001, −0.000, −0.002, and 0.000 over the four years of our sample.

Turning to abnormal volume, Fig. 6 demonstrates that, over the four years of our study, this value was comparatively higher for the premium group in year *t*-2 (median $ABVOL_{i,t}$

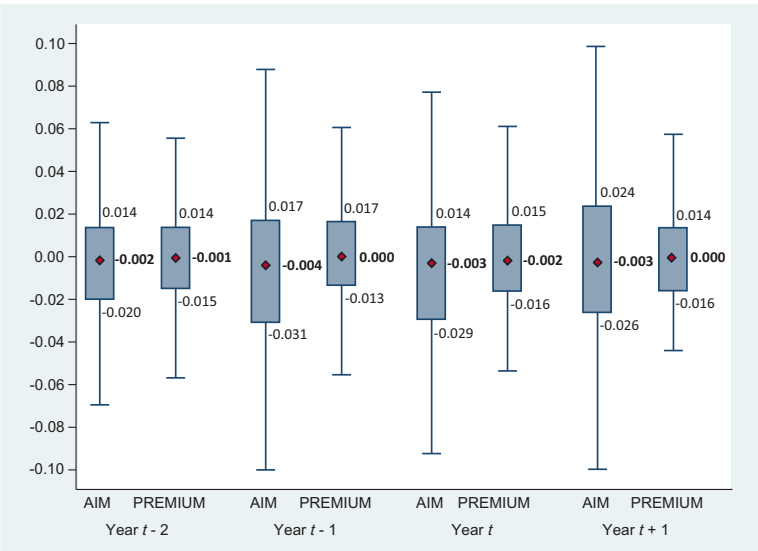

Fig. 5 Signed abnormal returns

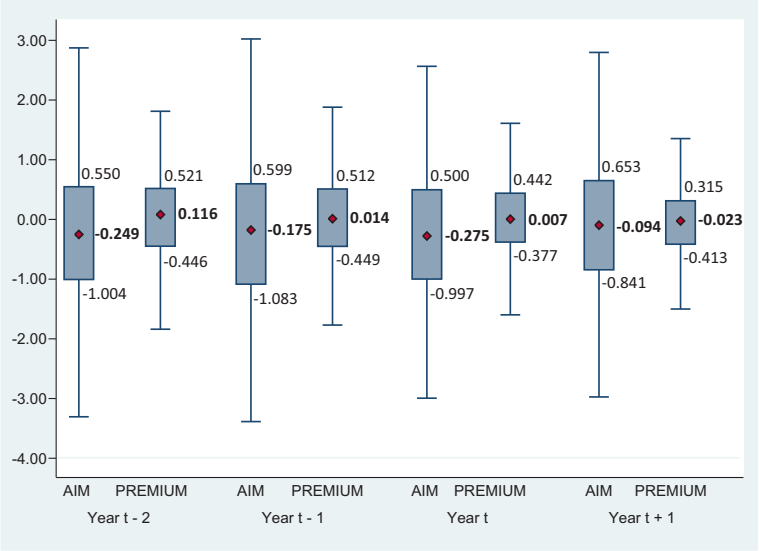

Fig. 6 Abnormal volume (*ABVOL*)

0.116) and remained stable, with a very small downward trend, in subsequent years (median $ABVOL_{i,t}$ is 0.014, 0.007, and − 0.023). For the AIM group, the median $ABVOL_{i,t}$ was −0.249, −0.175, −0.275, and − 0.094. Figure 6 also shows that AIM companies had a comparatively larger dispersion in abnormal volume, a finding consistent with the relatively high volatility that characterizes smaller firms. The combined evidence in Figs. 4 to 6 illustrates that the market's reaction on the filing date of the expanded auditor's report is very small in magnitude and has only slight variation over time for premium companies.

Panel A of Table 2 shows a test of overall differences in means for abnormal returns and volume before and after adoption of the new rules ($POST_{i,t} = 1$ and $POST_{i,t} = 0$) that was performed independently for AIM ($ADOPT_{i,t} = 0$) and premium companies ($ADOPT_{i,t} = 1$). For AIM companies, the differences in mean $ABRET_{i,t}$ and $ABVOL_{i,t}$ are statistically insignificant at conventional levels (diff in means $ABRET_{i,t} = 0.007$ and $ABVOL_{i,t} = 0.062$, $p > 0.10$). The differences in mean $ABRET_{i,t}$ and $ABVOL_{i,t}$ are also statistically insignificant for premium companies (diff in means $ABRET_{i,t} = 0.001$ and $ABVOL_{i,t} = -0.051$, $p > 0.10$). Panel A also reports descriptive statistics for the control variables used in our investors' reaction analyses. Most of these variables remained stable over time; the three exceptions were market value of equity ($LOGMKT_{i,t}$), which increased for both AIM (diff in means = 0.296, $p < 0.01$) and premium (diff in means = 0.281, $p < 0.10$) companies, market-to-book ratio ($MTB_{i,t}$), which increased for premium companies (diff in means = 0.916, $p < 0.01$), and leverage ($LEV_{i,t}$), which increased for AIM companies (diff in means = 0.014, $p < 0.10$). The time-related differences in market value and market-to-book ratio are somewhat expected given the rising trend in stock prices shown in Fig. 1: the FTSE 100 Index rose steadily from approximately 5300 in September 2011 to a maximum of over 7000 in April 2015 and closed at 6162 in September 2015.

Table 2 Analyses of market reaction to the filing of the expanded auditor's report

Panel A: Descriptive statistics

Variables	$ADOPT_{i,t} = 0, POST_{i,t} = 0$			$ADOPT_{i,t} = 0, POST_{i,t} = 1$			Diff. in Means
	Mean	Std. Dev.	Median	Mean	Std. Dev.	Median	
$ABRET_{i,t}$	0.043	0.064	0.019	0.050	0.113	0.023	0.007
$ABVOL_{i,t}$	−0.286	1.485	−0.208	−0.224	1.400	−0.175	0.062
$LOGMKT_{i,t}$	9.783	1.372	9.842	10.079	1.422	10.094	0.296***
$ROA_{i,t}$	−0.127	0.346	−0.003	−0.134	0.352	−0.008	−0.007
$LOSS_{i,t}$	0.505	0.500	1.000	0.517	0.500	1.000	0.012
$MTB_{i,t}$	2.522	4.164	1.386	2.547	3.690	1.651	0.025
$LEV_{i,t}$	0.056	0.116	0.000	0.070	0.137	0.001	0.014*
$LAG_{i,t}$	25.360	25.050	20.000	22.918	23.262	17.000	−2.442
$SALEVOL_{i,t}$	0.258	0.304	0.172	0.227	0.283	0.149	−0.031
$CHNI_{i,t}$	−0.001	0.227	0.003	0.000	0.230	0.004	0.001
$BETA_{i,t}$	0.274	0.598	0.176	0.325	0.670	0.238	0.051
$ABRET_EAR_{i,t}$	0.065	0.071	0.043	0.069	0.072	0.047	0.004
$BIG_{i,t}$	0.325	0.469	0.000	0.335	0.472	0.000	0.010
N. Obs.	656			656			1312

Variables	$ADOPT_{i,t} = 1, POST_{i,t} = 0$			$ADOPT_{i,t} = 1, POST_{i,t} = 1$			Diff. in Means
	Mean	Std. Dev.	Median	Mean	Std. Dev.	Median	
$ABRET_{i,t}$	0.023	0.028	0.014	0.024	0.029	0.015	0.001
$ABVOL_{i,t}$	−0.007	0.982	0.084	−0.058	0.957	−0.006	−0.051
$LOGMKT_{i,t}$	13.270	1.972	13.311	13.551	1.953	13.654	0.281*
$ROA_{i,t}$	0.048	0.154	0.058	0.038	0.144	0.052	−0.010
$LOSS_{i,t}$	0.119	0.324	0.000	0.168	0.374	0.000	0.049
$MTB_{i,t}$	2.497	3.409	1.793	3.413	4.221	2.362	0.916***
$LEV_{i,t}$	0.168	0.167	0.142	0.173	0.163	0.145	0.005
$LAG_{i,t}$	27.909	14.446	28.000	26.707	14.752	26.000	−1.202
$SALEVOL_{i,t}$	0.148	0.160	0.115	0.134	0.161	0.098	−0.014
$CHNI_{i,t}$	0.003	0.095	0.006	0.007	0.118	0.004	0.004
$BETA_{i,t}$	0.997	0.718	0.903	0.987	0.683	0.986	−0.010
$ABRET_EAR_{i,t}$	0.048	0.048	0.034	0.046	0.046	0.033	0.002
$BIG_{i,t}$	0.936	0.245	1.000	0.920	0.272	1.000	−0.016
N. Obs.	624			624			1248

Panel B: Difference-in-differences and pre-post analyses of abnormal returns

Variables	Dep. Var. = $ABRET_{i,t}$				
	(1)	(2)	(3)	(4)	(5)
$POST_{i,t}$	0.006	0.006	0.009	0.001	0.001
	(1.26)	(1.28)	(1.43)	(0.79)	(0.75)
$ADOPT_{i,t}$	−0.003	−0.003			
	(−0.74)	(−0.78)			
$POST*ADOPT_{i,t}$	−0.004	−0.004	−0.004		
	(−0.88)	(−0.87)	(−0.82)		
$LOGMKT_{i,t}$	−0.002***	−0.002***	−0.010	−0.002***	−0.002***
	(−2.64)	(−2.81)	(−1.58)	(−3.98)	(−4.27)
$ROA_{i,t}$	−0.006	−0.006	−0.010	−0.008	−0.008
	(−0.61)	(−0.61)	(−0.73)	(−1.01)	(−0.67)
$LOSS_{i,t}$	0.003	0.003	−0.009*	0.004	0.004
	(0.74)	(0.81)	(−1.66)	(1.18)	(1.13)
$MTB_{i,t}$	−0.000	−0.000	−0.000	−0.000	−0.000
	(−0.29)	(−0.30)	(−0.35)	(−0.23)	(−0.18)
$LEV_{i,t}$	0.017*	0.017	−0.013	0.017*	0.017**
	(1.65)	(1.60)	(−0.55)	(1.82)	(2.33)

Table 2 (continued)

$LAG_{i,t}$	−0.000***	−0.000***	−0.000***	−0.000***	−0.000***
	(−4.00)	(−4.37)	(−2.71)	(−4.42)	(−5.42)
$SALEVOL_{i,t}$	0.025	0.025	0.006	0.005	0.005
	(1.49)	(1.61)	(0.38)	(0.89)	(0.86)
$CHNI_{i,t}$	−0.002	−0.002	−0.005	0.022**	0.022*
	(−0.32)	(−0.29)	(−0.70)	(2.18)	(1.71)
$BETA_{i,t}$	0.004	0.004	−0.001	0.003*	0.003*
	(1.40)	(1.53)	(−0.25)	(1.73)	(1.75)
$ABRET_EAR_{i,t}$	0.375***	0.375***	0.346***	0.158***	0.158***
	(5.60)	(5.79)	(5.84)	(4.25)	(4.96)
$BIG_{i,t}$	−0.002	−0.002	0.009	0.006*	0.006**
	(−0.64)	(−0.72)	(1.22)	(1.75)	(2.00)
Intercept	0.029***	0.029***	0.133*	0.049***	0.049***
	(2.62)	(2.96)	(1.84)	(6.01)	(5.81)
Industry FE	*Included*	*Included*		*Included*	*Included*
Company FE			*Included*		
Bootstrap S.E.		*Yes*			*Yes*
N. Obs.	2560	2560	2560	1248	1248
Adj. R^2	0.18	0.18	0.12	0.18	0.18

Panel C: Difference-in-differences and pre-post analyses of abnormal volume

Variables	**Dep. Var. = $ABVOL_{i,t}$**				
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
$POST_{i,t}$	0.006	0.006	0.000	−0.068	−0.068
	(0.08)	(0.08)	(0.00)	(−1.30)	(−1.24)
$ADOPT_{i,t}$	−0.014	−0.014			
	(−0.14)	(−0.15)			
$POST*ADOPT_{i,t}$	−0.080	−0.080	−0.075		
	(−0.83)	(−0.84)	(−0.78)		
$LOGMKT_{i,t}$	0.090***	0.090***	0.134**	0.075***	0.075***
	(4.64)	(4.78)	(2.03)	(3.91)	(3.67)
$ROA_{i,t}$	0.128	0.128	−0.110	0.566**	0.566*
	(0.82)	(0.94)	(−0.45)	(2.27)	(1.69)
$LOSS_{i,t}$	−0.189**	−0.189**	−0.188*	−0.123	−0.123
	(−2.34)	(−2.44)	(−1.70)	(−1.04)	(−1.02)
$MTB_{i,t}$	0.003	0.003	−0.009	−0.001	−0.001
	(0.44)	(0.47)	(−1.11)	(−0.27)	(−0.26)
$LEV_{i,t}$	−0.016	−0.016	−0.455	0.193	0.193
	(−0.08)	(−0.09)	(−1.12)	(1.13)	(1.13)
$LAG_{i,t}$	−0.009***	−0.009***	−0.010***	−0.010***	−0.010***
	(−6.55)	(−6.95)	(−4.37)	(−4.64)	(−4.52)
$SALEVOL_{i,t}$	0.174	0.174	0.247	0.027	0.027
	(1.21)	(1.36)	(0.77)	(0.17)	(0.15)
$CHNI_{i,t}$	0.075	0.075	0.050	−0.068	−0.068
	(0.39)	(0.39)	(0.23)	(−0.20)	(−0.16)

Table 2 (continued)

$BETA_{i,t}$	0.059	0.059	0.000	0.028	0.028
	(1.41)	(1.41)	(0.00)	(0.51)	(0.53)
$ABRET_EAR_{i,t}$	2.892***	2.892***	2.890***	1.906***	1.906***
	(6.36)	(6.41)	(5.88)	(3.32)	(3.58)
$BIG_{i,t}$	−0.110	−0.110	−0.027	−0.042	−0.042
	(−1.39)	(−1.54)	(−0.15)	(−0.25)	(−0.25)
Intercept	−1.313***	−1.313***	−1.490*	−1.160*	−1.160**
	(−3.91)	(−4.26)	(−1.90)	(−1.79)	(−2.23)
Industry FE	*Included*	*Included*		*Included*	*Included*
Company FE			*Included*		
Bootstrap S.E.		*Yes*			*Yes*
N. Obs.	2560	2560	2560	1248	1248
Adj. R^2	0.0609	0.0609	0.0325	0.0554	0.0554

Panel A of this table includes descriptive statistics for the dependent and independent variables for the analyses of investors' reaction to the expanded auditor's report filing, using data for premium ($ADOPT_{i,t} = 1$) and AIM companies ($ADOPT_{i,t} = 0$) for two years before ($POST_{i,t} = 0$) and two years after ($POST_{i,t} = 1$) the regulatory cut-off date (September 2013). Panels B and C, Columns (1) to (3), show coefficients and t-statistics for our difference-in-differences model (Eq. 1). Column (1) shows the results including industry fixed effects, Column (2) the results including industry fixed effects and calculating standard errors using the bootstrap method with 1000 replications, and Column (3) the results including company fixed effects and excluding the $ADOPT_{i,t}$ intercept and industry fixed effects. Panels B and C, Columns (4) and (5), show the pre-post model for premium companies only. Column (4) shows the results including industry fixed effects, and Column (5) the results including industry fixed effects and calculating standard errors using the bootstrap method with 1000 replications. The continuous variables are winsorized at 1 and 99%, except for $ABRET_{i,t}$ and $ABVOL_{i,t}$. All variables are as described in Appendix 1. ***, **, * indicate statistical significance from two-tailed tests at 0.01, 0.05, and 0.1, respectively. All *t*-statistics are clustered by company

Panel B of Table 2 shows the results using cumulative absolute abnormal returns $ABRET_{i,t}$ as the dependent variable. Columns (1) through (3) show the coefficients and t-statistics for our difference-in-differences model (Eq. 1): Column (1) provides the results including industry fixed effects, Column (2) the results including industry fixed effects and calculating standard errors using the bootstrap method, and Column (3) the results including company fixed effects and excluding the $ADOPT_{i,t}$ intercept and industry fixed effects. The statistically insignificant difference-in-differences effect in Column (1) ($POST*ADOPT_{i,t} = -0.004$, $p > 0.10$) is robust to computing standard errors using the bootstrap method (Column 2) and to including company fixed effects (Column 3). The next two columns show the pre-post model for premium companies only: Column (4) presents the results including industry fixed effects, and Column (5) the results including these effects and calculating standard errors using the bootstrap method. Columns (4) and (5) confirm the trend apparent in our descriptive statistics, in which there are statistically insignificant coefficients for the $POST_{i,t}$ indicator (0.001, $p > 0.10$).

Panel C of Table 2 shows the results using abnormal volume $ABVOL_{i,t}$ as the dependent variable. The statistically insignificant difference-in-differences effect seen in Column (1) ($POST*ADOPT_{i,t} = -0.080$, $p > 0.10$) is robust to computing standard errors using the bootstrap method (Column 2) and to the inclusion of company fixed effects (Column 3).

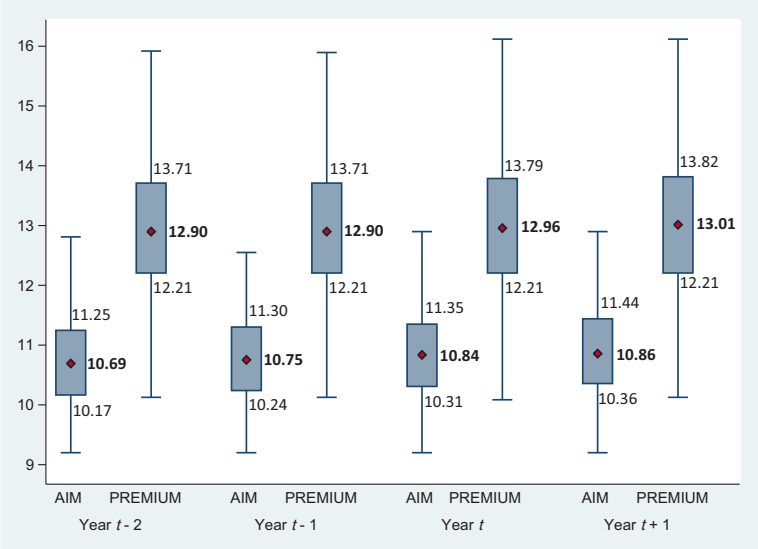

Fig. 7 Logarithm of audit fees (*FEES*)

In terms of the pre-post effect for premium companies, Columns (4) and (5) confirm the trend shown in our descriptive statistics, in which there is a statistically insignificant coefficient for the $POST_{i,t}$ indicator (−0.068, $p > 0.10$).[26] We conclude that our results are in line with the findings presented in the FRC post-implementation reviews. In the first-year review (FRC 2015, p. 60), investors voiced a preference that the expanded auditor's report be made available in the context of the preliminary announcement (i.e., on earnings release date), and some mentioned that the publication of the annual report does not typically "move the market."[27] In the FRC second-year review (FRC 2016, p. 8), investors reported that they were "still learning to decode and evaluate the language used by auditors in their reports."

4.3 Audit fees

Figure 7 shows that premium companies pay considerably higher audit fees (represented in the figure in log units). This differential is not, however, necessarily an issue for our study, since fees show a stable trend for both groups; in other words, the counterfactual levels of treated and nontreated groups differ, but their time variation appears to be similar.

[26] We also examine whether the auditor's report is potentially more informative for investors when companies have large earnings surprises. Arguably, there is comparatively higher investor attention in these cases. For premium companies, we estimate our pre-post model using the subsample with large (i.e., above the median) absolute returns on the earnings announcement day. For this group of companies, we also do not find evidence of a pre-post adoption effect on our market reaction variables.

[27] The FRC argues that this is a possibility in the discussion paper "Auditors and Preliminary Announcement" in April 2017 (FRC 2017).

The median $FEES_{i,t}$ for premium companies is 12.90, 12.90, 12.96, and 13.01, respectively, representing a range of approximately 400,000 to 447,000 British pounds.[28]

Turning now to Table 3, Panel A shows a test of overall differences in means for audit fees before and after the adoption of the new rules that was performed independently for AIM and premium companies. For AIM companies, fees increased on average approximately 8% (diff in means $FEES_{i,t} = 0.081$, $p < 0.10$), representing some 4000 pounds. By contrast, there was for premium companies a statistically insignificant increase in fees (diff in means $FEES_{i,t} = 0.060$, $p > 0.10$). Panel A also shows that most control variables remain stable over time, though assets ($SIZE_{i,t}$) increase and sales volatility ($SALEVOL_{i,t}$) decreases for AIM companies, while $LOSS_{i,t}$ and going-concern opinions ($GCO_{i,t}$) are slightly more common among premium companies. Panel B in Table 3 shows the results for our regression analyses of audit fees. They reveal a statistically insignificant difference-in-differences effect ($POST*ADOPT_{i,t} = -0.012$, $p > 0.10$) that is robust to computing standard errors using the bootstrap method (Column 2) and to the inclusion of company fixed effects (Column 3). In Column (3), the coefficient on the $POST_{i,t}$ indicator (0.036, $p > 0.05$) shows, after controlling for time-invariant client characteristics, a small increase in fees ($e^{\wedge 0.036} =$ 3.67%) for AIM companies (i.e., when $ADOPT_{i,t} = 0$). Also, in Column (1), the coefficient on the $ADOPT_{i,t}$ indicator (0.222, $p > 0.01$) shows that, on average, premium companies pay higher fees than AIM companies. Taking into account the pre-post effect for premium companies alone, Columns (3) and (4) confirm the trend found in our descriptive statistics, in that the coefficient for the $POST_{i,t}$ indicator (0.000, $p > 0.10$) is statistically insignificant. Our findings generally support the expectations of the FRC and of investors that the incremental costs associated with the expanded auditor's report would not be significant in relation to total audit costs.

4.4 Audit quality

Figure 8 shows that premium companies have comparatively lower absolute discretionary accruals. However, absolute discretionary accruals remain stable for both AIM and premium companies throughout the years in our sample. The median $DACCR_{i,t}$ for premium companies is 0.034, 0.029, 0.031, and 0.035, respectively. Turning now to Table 4, Panel A shows a test of overall differences in means for absolute discretionary accruals before and after the adoption of the new rules that was performed independently for AIM and premium companies. For both groups, the change in absolute discretionary accruals is statistically insignificant ($DACCR_{i,t} = 0.001$ in each case, $p > 0.10$).

Panel B in Table 4 shows the regression results for our analyses of audit quality, revealing a statistically insignificant difference-in-differences effect in Column (1) ($POST*ADOPT_{i,t} = -0.001$, $p > 0.10$) that is robust to computing standard errors using the bootstrap method (Column 2) and to the inclusion of company fixed effects (Column 3). In terms of the pre-post effect for premium companies alone, the coefficient for the $POST_{i,t}$ indicator in Columns (4) and (5) (−0.000, $p > 0.10$) is statistically insignificant. Our results are consistent with the FRC's expressed intention to provide greater transparency without affecting the underlying audit work (at least in the short run).

[28] A visual inspection of figures 4 through 7 suggests that the parallel trend assumption requirement for difference-in-differences holds for all our dependent variables in the two years before the expanded auditor's report requirements.

Table 3 Audit fee analyses

Panel A: Descriptive statistics

Variables	$ADOPT_{i,t} = 0$, $POST_{i,t} = 0$			$ADOPT_{i,t} = 0$, $POST_{i,t} = 1$			Diff. in Means
	Mean	Std. Dev.	Median	Mean	Std. Dev.	Median	
$FEES_{i,t}$	10.757	0.787	10.723	10.838	0.779	10.840	0.081*
$SIZE_{i,t}$	9.772	1.410	9.805	9.930	1.405	10.007	0.158*
$ROA_{i,t}$	−0.142	0.360	−0.009	−0.135	0.342	−0.011	0.007
$LOSS_{i,t}$	0.515	0.500	1.000	0.524	0.500	1.000	0.009
$MTB_{i,t}$	2.458	4.104	1.402	2.522	3.646	1.668	0.064
$LEV_{i,t}$	0.055	0.114	0.000	0.066	0.133	0.000	0.011
$CFO_{i,t}$	−0.060	0.279	0.015	−0.053	0.263	0.016	0.007
$SALEVOL_{i,t}$	0.256	0.303	0.168	0.223	0.275	0.146	−0.033*
$BIG_{i,t}$	0.326	0.469	0.000	0.335	0.472	0.000	0.009
$INV_{i,t}$	0.065	0.108	0.009	0.063	0.109	0.009	−0.002
$REC_{i,t}$	0.173	0.153	0.134	0.170	0.149	0.135	−0.003
$FORSALE_{i,t}$	0.213	0.313	0.000	0.199	0.309	0.000	−0.014
$BUSY_{i,t}$	0.639	0.481	1.000	0.640	0.480	1.000	0.001
$GCO_{i,t}$	0.076	0.266	0.000	0.068	0.252	0.000	−0.008
$USLIST_{i,t}$	0.006	0.074	0.000	0.004	0.064	0.000	−0.002
$NSEG_{i,t}$	0.557	0.618	0.000	0.499	0.612	0.000	−0.058
N. Obs.	720			720			1440

Variables	$ADOPT_{i,t} = 1$, $POST_{i,t} = 0$			$ADOPT_{i,t} = 1$, $POST_{i,t} = 1$			Diff in Means
	Mean	Std. Dev.	Median	Mean	Std. Dev.	Median	
$FEES_{i,t}$	13.024	1.288	12.899	13.084	1.297	12.981	0.060
$SIZE_{i,t}$	13.475	1.865	13.410	13.546	1.855	13.501	0.071
$ROA_{i,t}$	0.049	0.154	0.057	0.039	0.142	0.053	−0.010
$LOSS_{i,t}$	0.117	0.322	0.000	0.160	0.367	0.000	0.043*
$MTB_{i,t}$	2.526	3.409	1.807	3.414	4.140	2.399	0.888***
$LEV_{i,t}$	0.168	0.166	0.142	0.172	0.161	0.146	0.004
$CFO_{i,t}$	0.091	0.112	0.090	0.088	0.102	0.084	−0.003
$SALEVOL_{i,t}$	0.146	0.141	0.115	0.132	0.143	0.098	−0.014
$BIG_{i,t}$	0.932	0.251	1.000	0.931	0.254	1.000	−0.001
$INV_{i,t}$	0.110	0.134	0.070	0.107	0.135	0.061	−0.003
$REC_{i,t}$	0.161	0.131	0.134	0.166	0.133	0.142	0.005
$FORSALE_{i,t}$	0.404	0.365	0.343	0.401	0.360	0.346	−0.003
$BUSY_{i,t}$	0.764	0.425	1.000	0.766	0.424	1.000	0.002
$GCO_{i,t}$	0.008	0.091	0.000	0.026	0.160	0.000	0.018*
$USLIST_{i,t}$	0.053	0.224	0.000	0.053	0.224	0.000	0.000
$NSEG_{i,t}$	0.942	0.655	1.099	0.957	0.680	1.099	0.015
N. Obs.	606			606			1212

Panel B: Difference-in-differences and pre-post analyses

Variables	Dep. Var. = $FEES_{i,t}$				
	(1)	(2)	(3)	(4)	(5)
$POST_{i,t}$	0.019	0.019	0.036**	0.000	0.000
	(1.19)	(0.76)	(2.50)	(0.00)	(0.00)
$ADOPT_{i,t}$	0.222***	0.222***			
	(3.76)	(5.75)			
$POST*ADOPT_{i,t}$	−0.012	−0.012	0.002		
	(−0.57)	(−0.29)	(0.12)		
$SIZE_{i,t}$	0.536***	0.536***	0.294***	0.597***	0.597***
	(31.57)	(56.82)	(9.97)	(26.98)	(47.47)
$ROA_{i,t}$	−0.255***	−0.255***	−0.128***	−0.362	−0.362
	(−3.59)	(−3.73)	(−2.93)	(−1.58)	(−1.58)

Table 3 (continued)

$LOSS_{i,t}$	0.073*	0.073**	0.010	0.083	0.083
	(1.88)	(2.39)	(0.55)	(1.17)	(1.39)
$MTB_{i,t}$	0.004	0.004	−0.001	0.000	0.000
	(0.95)	(1.41)	(−0.44)	(0.00)	(0.01)
$LEV_{i,t}$	0.004	0.004	0.084	−0.004	−0.004
	(0.03)	(0.05)	(1.02)	(−0.02)	(−0.04)
$CFO_{i,t}$	−0.126	−0.126	−0.036	0.098	0.098
	(−1.33)	(−1.59)	(−0.54)	(0.36)	(0.42)
$SALEVOL_{i,t}$	0.155**	0.155***	−0.018	0.279	0.279*
	(1.96)	(3.03)	(−0.26)	(1.15)	(1.65)
$BIG_{i,t}$	0.095*	0.095***	0.159**	0.059	0.059
	(1.89)	(3.27)	(2.41)	(0.43)	(0.72)
$INV_{i,t}$	−0.279	−0.279**	0.716**	−0.070	−0.070
	(−1.31)	(−2.44)	(2.24)	(−0.26)	(−0.47)
$REC_{i,t}$	0.879***	0.879***	0.125	1.291***	1.291***
	(6.33)	(10.88)	(1.02)	(5.11)	(8.93)
$FORSALE_{i,t}$	0.490***	0.490***	0.104***	0.693***	0.693***
	(8.30)	(13.98)	(2.88)	(8.47)	(13.61)
$BUSY_{i,t}$	0.025	0.025	0.053	0.052	0.052
	(0.60)	(1.10)	(0.66)	(0.73)	(1.38)
$GCO_{i,t}$	0.115*	0.115**	0.061	0.475***	0.475***
	(1.71)	(2.26)	(1.52)	(2.81)	(2.93)
$USLIST_{i,t}$	0.503***	0.503***	−0.837***	0.355***	0.355***
	(4.21)	(7.93)	(−18.37)	(2.83)	(5.27)
$NSEG_{i,t}$	0.112***	0.112***	0.013	0.148***	0.148***
	(3.69)	(6.25)	(0.64)	(3.50)	(6.09)
Intercept	4.527***	4.527***	8.173***	3.577***	3.577***
	(18.01)	(31.71)	(23.25)	(13.71)	(16.60)
Industry FE	*Included*	*Included*		*Included*	*Included*
Company FE			*Included*		
Bootstrap S.E.		*Yes*			*Yes*
N. Obs.	2652	2652	2652	1212	1212
Adj. R^2	0.89	0.89	0.24	0.84	0.84

Panel A of this table includes descriptive statistics for the dependent and independent variables for the analyses of audit fees using data for premium ($ADOPT_{i,t} = 1$) and AIM companies ($ADOPT_{i,t} = 0$) for two years before ($POST_{i,t} = 0$) and two years after ($POST_{i,t} = 1$) the regulatory cut-off date (September 2013). Panels B, Columns (1) to (3), show coefficients and t-statistics for our difference-in-differences model (Eq. 1). Column (1) shows the results including industry fixed effects, Column (2) the results including industry fixed effects and calculating standard errors using the bootstrap method with 1000 replications, and Column (3) the results including company fixed effects and excluding the $ADOPT_{i,t}$ intercept and industry fixed effects. Panels B, Columns (4) and (5), show the pre-post model for premium companies only. Column (4) shows the results including industry fixed effects, and Column (5) the results including industry fixed effects and calculating standard errors using the bootstrap method with 1000 replications. The continuous variables are winsorized at 1 and 99%. All variables are as described in Appendix 1. ***, **, * indicate statistical significance from two-tailed tests at 0.01, 0.05, and 0.1, respectively. All *t*-statistics are clustered by company

5 Additional and sensitivity analyses

5.1 Results of additional cross-sectional analyses

Table 5, Panel A, shows descriptive statistics relating to the content of the auditor's report calculated for premium companies in the adoption years t and $t+1$ using the sample described in Table 1. Under the new rules, the reports average 2532 words, with a median of 2374 words or approximately three pages. The risks section typically includes four risks (mean 3.97 and median 4.00) but only one that is not mentioned in the audit committee report (mean 0.78 and median 1.00). The materiality scaled by total assets averages 0.64%, with a median of 0.50 or approximately 0.5% of total assets.

Panel B of Table 5 shows the results of estimating our pre-post model (Eq. 2) by interacting the $POST_{i,t}$ indicator with four partition variables indicating long reports $LENGTH(1/0)_{i,t}$, large number of risks $NRISKS(1/0)_{i,t}$, large number of unique auditor risks $NRISKSAUD(1/0)_{i,t}$, and large materiality $MATPERC(1/0)_{i,t}$. Columns (1) through (4) use cumulative absolute abnormal returns as the dependent variable, while Columns (5) through (8) use cumulative abnormal volume as the dependent variable. We find a positive coefficient only for the interaction $POST*LENGTH(1/0)_{i,t}$ (0.187, $p<0.10$), a result that indicates an increase in volume in the post-adoption period for companies using the long auditor's report. However, we conclude that this is an isolated finding

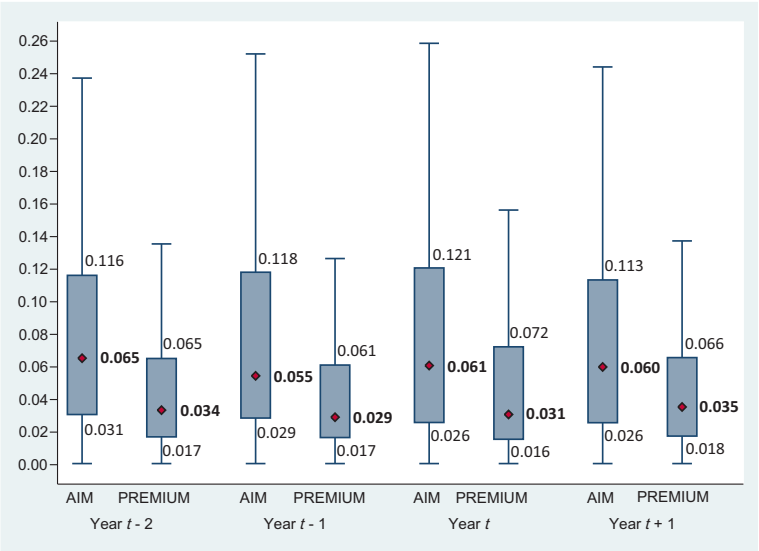

Fig. 8 Absolute discretionary accruals (*DACCR*)

Table 4 Audit quality analyses

Panel A: Descriptive statistics

Variables	$ADOPT_{i,t} = 0$, $POST_{i,t} = 0$			$ADOPT_{i,t} = 0$, $POST_{i,t} = 1$			Diff. in Means
	Mean	Std. Dev.	Median	Mean	Std. Dev.	Median	
$DACCR_{i,t}$	0.088	0.088	0.062	0.089	0.090	0.061	0.001
$SIZE_{i,t}$	9.671	1.369	9.664	9.828	1.379	9.927	0.157
$ROA_{i,t}$	−0.151	0.367	−0.024	−0.160	0.367	−0.025	−0.009
$LOSS_{i,t}$	0.544	0.498	1.000	0.554	0.497	1.000	0.010
$MTB_{i,t}$	2.626	4.151	1.543	2.507	3.557	1.692	−0.119
$LEV_{i,t}$	0.046	0.103	0.000	0.060	0.127	0.000	0.014
$CFO_{i,t}$	−0.064	0.279	0.002	−0.063	0.274	0.007	0.001
$SALEVOL_{i,t}$	0.259	0.315	0.163	0.226	0.287	0.141	−0.033
$BIG_{i,t}$	0.296	0.457	0.000	0.318	0.466	0.000	0.022
$LAGACCR_{i,t}$	−0.044	0.183	−0.005	−0.042	0.176	−0.008	0.002
N. Obs.	592			592			1184

Variables	ADOPTi,t = 1, POSTi,t = 0			$ADOPT_{i,t} = 1$, $POST_{i,t} = 1$			Diff. in Means
	Mean	Std. Dev.	Median	Mean	Std. Dev.	Median	
$DACCR_{i,t}$	0.049	0.056	0.031	0.050	0.055	0.034	0.001
$SIZE_{i,t}$	13.390	1.901	13.241	13.453	1.909	13.339	0.063
$ROA_{i,t}$	0.047	0.177	0.059	0.037	0.156	0.055	−0.010
$LOSS_{i,t}$	0.106	0.308	0.000	0.163	0.370	0.000	0.057*
$MTB_{i,t}$	2.563	3.230	2.021	3.427	3.974	2.432	0.864***
$LEV_{i,t}$	0.167	0.173	0.128	0.173	0.167	0.143	0.006
$CFO_{i,t}$	0.090	0.125	0.091	0.089	0.113	0.088	−0.001
$SALEVOL_{i,t}$	0.150	0.158	0.112	0.137	0.163	0.093	−0.013
$BIG_{i,t}$	0.927	0.261	1.000	0.913	0.282	1.000	−0.014
$LAGACCR_{i,t}$	0.006	0.071	0.007	−0.006	0.114	0.007	−0.012
N. Obs.	436			436			872

Panel B: Difference-in-differences and pre-post analyses

Variables	Dep. Var. = $DACCR_{i,t}$				
	(1)	(2)	(3)	(4)	(5)
$POST_{i,t}$	0.001	0.001	−0.001	−0.000	−0.000
	(0.20)	(0.21)	(−0.19)	(−0.05)	(−0.05)
$ADOPT_{i,t}$	0.008	0.008			
	(1.45)	(1.55)			
$POST*ADOPT_{i,t}$	−0.001	−0.001	0.001		
	(−0.09)	(−0.10)	(0.19)		
$SIZE_{i,t}$	−0.005***	−0.005***	0.001	−0.004**	−0.004***
	(−3.81)	(−4.21)	(0.07)	(−2.56)	(−2.97)
$ROA_{i,t}$	−0.138***	−0.138***	−0.124***	−0.157***	−0.157***
	(−7.61)	(−7.72)	(−5.94)	(−5.33)	(−4.03)
$LOSS_{i,t}$	−0.009*	−0.009*	−0.010	0.004	0.004
	(−1.74)	(−1.87)	(−1.51)	(0.58)	(0.58)
$MTB_{i,t}$	0.000	0.000	0.000	0.001	0.001
	(0.55)	(0.62)	(0.31)	(1.31)	(1.44)
$LEV_{i,t}$	−0.005	−0.005	−0.058*	−0.009	−0.009
	(−0.37)	(−0.38)	(−1.73)	(−0.47)	(−0.58)
$CFO_{i,t}$	0.037	0.037	0.060	0.139***	0.139***
	(1.54)	(1.56)	(1.56)	(3.25)	(2.94)

Table 4 (continued)

$SALEVOL_{i,t}$	0.014	0.014	−0.036	0.070***	0.070***
	(1.46)	(1.56)	(−1.58)	(3.33)	(2.99)
$BIG_{i,t}$	−0.006	−0.006	−0.006	0.000	0.000
	(−1.40)	(−1.50)	(−0.60)	(0.04)	(0.04)
$LAGACCR_{i,t}$	−0.026*	−0.026	−0.013	−0.023	−0.023
	(−1.67)	(−1.41)	(−0.77)	(−0.65)	(−0.51)
Intercept	0.126***	0.126***	0.076	0.095***	0.095***
	(7.68)	(8.60)	(0.81)	(4.52)	(5.03)
Industry FE	*Included*	*Included*		*Included*	*Included*
Company FE			*Included*		
Bootstrap S.E.		*Yes*			*Yes*
N. Obs.	2056	2056	2056	872	872
Adj. R2	0.32	0.32	0.11	0.25	0.25

Panel A of this table includes descriptive statistics for the dependent and independent variables for the analyses of audit quality using data for premium ($ADOPT_{i,t} = 1$) and AIM companies ($ADOPT_{i,t} = 0$) for two years before ($POST_{i,t} = 0$) and two years after ($POST_{i,t} = 1$) the regulatory cut-off date (September 2013). Panels B, Columns (1) to (3), show coefficients and t-statistics for our difference-in-differences model (Eq. 1). Column (1) shows the results including industry fixed effects, Column (2) the results including industry fixed effects and calculating standard errors using the bootstrap method with 1000 replications, and Column (3) the results including company fixed effects and excluding the $ADOPT_{i,t}$ intercept and industry fixed effects. Panels B, Columns (4) and (5), show the pre-post model for premium companies only. Column (4) shows the results including industry fixed effects, and Column (5) the results including industry fixed effects and calculating standard errors using the bootstrap method with 1000 replications. The continuous variables are winsorized at 1 and 99%. All variables are as described in Appendix 1. ***, **, * indicate statistical significance from two-tailed tests at 0.01, 0.05, and 0.1, respectively. All *t*-statistics are clustered by company

with low statistical power.[29] Once again, the lack of results regarding investors' reaction to variation in the auditor's report content is consistent with the findings of the FRC post-implementation reviews. Thus, in the first-year review, investors criticized the lack of granularity in the risk reporting (i.e., they found the explanations overly generic or abstract) (FRC 2015, p. 5), and both year reviews concluded that investors were dissatisfied with the explanations with regard to auditors' application of materiality (e.g., benchmark and level) (FRC 2015, p. 5; FRC 2016, p. 5). Moreover, the second-year review concluded that the expanded auditor's reports lack dynamic analyses of changes over time (FRC 2016, p. 4). In sum, it appeared somewhat difficult to distinguish the incremental information in these reports.

Panel C of Table 5 shows the cross-sectional analyses using audit fees as the dependent variable. We find no evidence that adopting firms characterized by long reports, a large number of risks, including unique auditor risks, and large materiality experienced an incremental pre-post change in audit fees (i.e., the coefficients on the interaction variables are statistically insignificant, $p > 0.10$). We do note, however, that firms with long reports $LENGTH(1/0)_{i,t}$ and large numbers of risks $NRISKS(1/0)_{i,t}$ pay comparatively higher fees, the statistically significant ($p < 0.01$) coefficients on these variables (i.e., the main effects) being

[29] In additional untabulated analyses, this coefficient, after winsorizing abnormal volume at the top and bottom 99 and 1%, becomes statistically insignificant ($p > 0.10$).

Table 5 Cross-sectional analyses taking into account variation in the content of the expanded auditor's report

Panel A: Content of the auditor's report

Variable	Mean	Std. Dev.	Median
Natural logarithm of number of words	2532	810	2374
Number of risks	3.97	1.47	4.00
Number of unique risks	0.78	1.16	1.00
Percentage materiality scaled by total assets	0.64	0.46	0.50
N. Obs.	1248		

Panel B: Content of the auditor's report and investors' reaction

Variables	Dep. Var. = $ABRET_{i,t}$				Dep. Var. = $ABVOL_{i,t}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$POST_{i,t}$	0.002	0.001	0.001	0.000	−0.161*	−0.096	−0.092	−0.143**
	(1.08)	(0.57)	(0.74)	(0.25)	(−1.90)	(−1.30)	(−1.35)	(−2.13)
$LENGTH(1/0)_{i,t}$	0.003				−0.086			
	(1.48)				(−1.09)			
$POST*LENGTH(1/0)_{i,t}$	−0.002				0.187*			
	(−0.85)				(1.79)			
$NRISKS_{i,t}$		0.002				−0.058		
		(0.77)				(−0.74)		
$POST*NRISKS(1/0)_{i,t}$		0.000				0.072		
		(0.08)				(0.76)		
$NRISKSAUD(1/0)_{i,t}$			0.002				0.038	
			(0.74)				(0.48)	
$POST*NRISKSAUD(1/0)_{i,t}$			−0.001				0.060	
			(−0.20)				(0.58)	
$MATPERC(1/0)_{i,t}$				0.002				−0.123
				(1.01)				(−1.63)
$POST*MATPERC(1/0)_{i,t}$				0.001				0.148
				(0.53)				(1.44)
Intercept	0.050***	0.050***	0.049***	0.049***	−1.101*	−1.158*	−1.147*	−1.105*
	(5.74)	(5.65)	(5.92)	(5.73)	(−1.69)	(−1.80)	(−1.76)	(−1.71)
Controls	*Included*	*Included*	*Included*	*Included*	*Included*	*Included*	*Included*	*Included*
Industry F.E.	*Included*	*Included*	*Included*	*Included*	*Included*	*Included*	*Included*	*Included*
N. Obs.	1248	1248	1248	1248	1248	1248	1248	1248
Adj. R^2	0.11	0.11	0.11	0.12	0.05	0.05	0.05	0.05

Panel C: Content of the auditor's report and audit fees

Variables	Dep. Var. = $FEES_{i,t}$			
	(1)	(2)	(3)	(4)
$POST_{i,t}$	−0.001	−0.015	−0.010	−0.020
	(−0.07)	(−0.81)	(−0.52)	(−1.08)
$LENGTH(1/0)_{i,t}$	0.237***			
	(4.00)			
$POST*LENGTH(1/0)_{i,t}$	0.012			
	(0.44)			
$NRISKS_{i,t}$		0.175***		
		(2.77)		
$POST*NRISKS(1/0)_{i,t}$		0.044		
		(1.62)		
$NRISKSAUD(1/0)_{i,t}$			0.022	
			(0.39)	
$POST*NRISKSAUD(1/0)_{i,t}$			0.025	
			(0.92)	

Table 5 (continued)

MATPERC(1/0)$_{i,t}$				−0.051
				(−0.86)
POST*MATPERC(1/0)$_{i,t}$				0.039
				(1.41)
Intercept	3.761***	3.674***	3.585***	3.618***
	(13.51)	(12.99)	(13.75)	(13.41)
Controls	*Included*	*Included*	*Included*	*Included*
Industry F.E.	*Included*	*Included*	*Included*	*Included*
N. Obs.	1212	1212	1212	1212
Adj. R^2	0.85	0.84	0.84	0.84

Panel D: Content of the auditor's report and audit quality

Variable	**Dep. Var. = DACCR$_{i,t}$**			
	(1)	**(2)**	**(3)**	**(4)**
POST$_{i,t}$	0.002	0.002	0.005	−0.004
	(0.47)	(0.40)	(1.34)	(−0.95)
LENGTH(1/0)$_{i,t}$	−0.000			
	(−0.04)			
POST*LENGTH(1/0)$_{i,t}$	−0.006			
	(−0.88)			
NRISKS$_{i,t}$		0.003		
		(0.56)		
POST*NRISKS(1/0)$_{i,t}$		−0.005		
		(−0.73)		
NRISKSAUD(1/0)$_{i,t}$			0.009	
			(1.60)	
POST*NRISKSAUD(1/0)$_{i,t}$			−0.012	
			(−1.63)	
MATPERC(1/0)$_{i,t}$				−0.001
				(−0.21)
POST*MATPERC(1/0)$_{i,t}$				0.007
				(1.13)
Intercept	0.091***	0.095***	0.091***	0.094***
	(4.20)	(4.44)	(4.40)	(4.28)
Controls	*Included*	*Included*	*Included*	*Included*
Industry F.E.	*Included*	*Included*	*Included*	*Included*
N. Obs.	872	872	872	872
Adj. R^2	0.25	0.25	0.25	0.25

Panel A of this table includes descriptive statistics for the average auditor's report length (natural logarithm of the number of words), number of risks, number of unique risks (those not mentioned in the audit committee's report), and percentage materiality (materiality in British pounds scaled by total assets*100) in years t and $t+1$, using the sample of premium companies with available data for our analyses of investors' reaction. Panels B to D show the results of our cross-sectional analyses interacting our $POST_{i,t}$ 2013 variable with indicator variables equal to one when average t and $t+1$ length, number of risks, number of unique risks, and percentage materiality are greater than the median of the distribution and equal to zero otherwise. All models in Panels B, C, and D include the same control variables as in Tables 2, 3, and 4, respectively. The continuous variables, except for $ABRET_{i,t}$ and $ABVOL_{i,t}$, are winsorized at 1 and 99%. All variables are as described in Appendix 1. ***, **, * indicate statistical significance from two-tailed tests at 0.01, 0.05, and 0.1, respectively. All t-statistics are clustered by company

0.237 and 0.175. This evidence suggests a positive association between the auditor's discussion of risks and his or her effort, at least to the extent that audit fees capture that effort. Nevertheless, *disclosing* the risks does not itself cause a *change* in audit fees or quality.[30]

Panel D of Table 5 shows the cross-sectional analyses using absolute discretionary accruals as the dependent variable. We find negligible evidence that the adopting firms characterized by long reports, a large number of risks, including unique auditor risks, and large materiality experienced an incremental pre-post change in absolute discretionary accruals (i.e., the coefficients on the interaction variables are statistically insignificant, $p > 0.10$).[31]

5.2 Related concurrent studies

Lennox et al. (2017) have examined investors' reaction to the auditor's report, focusing on the effect of the risk disclosures. Using short-window analyses (i.e., signed cumulative abnormal returns around the filing date), their study demonstrates that investors do not find the report's risk disclosures informative. Based on long-window analyses (i.e., levels equity valuation models), these researchers have concluded that the report's risk disclosures, while they capture uncertainty and are associated with client's risk, lack incremental information content. The sample used consists of premium companies in the two years surrounding the adoption of the new rules (t, $t+1$).[32] Lennox et al. (2017) additionally have analyzed changes in audit fees using a difference-in-differences analysis with AIM companies serving as a control group and find no significant change following the introduction of expanded auditor's report. Our inferences are consistent with these researchers' conclusions regarding the informativeness of the auditor's report and changes in audit fees and complement them by (1) providing triangulating evidence from several proxies for investors' reaction, (2)

[30] The link between materiality, audit risk, and fees is a complex issue. There could be several reasons why the indicator for high percentage materiality threshold is not significantly associated with audit fees, after controlling for other factors. We find univariate (untabulated) negative and statistically significant associations between (a) the natural logarithm of audit fees and percentage materiality (correlation = −0.327 $p < 0.01$), and (b) the number of risks in the expanded auditor's report and percentage materiality (correlation = −0.235 $p < 0.01$). However, the cross-sectional variation in materiality appears to be mechanically subsumed by variation in other client characteristics, such as size, profitability, etc.

[31] We repeated these analyses using a 75th percentile cut-off in defining the indicator variables for long reports, large number of risks, large number of unique auditor risks, and large materiality. These alternative definitions aim to capture significant or more extreme departures than a 50th percentile cut-off. We find (1) a negative and statistically significant coefficient ($p < 0.05$) for the $LENGTH(1/0)_{i,t}$ interaction in the absolute abnormal returns model, (2) a positive and weakly significant coefficient ($p < 0.10$) for the $NRISKS(1/0)_{i,t}$ interaction in the audit fees model, (3) a negatively and weakly significant coefficient ($p < 0.10$) for the $LENGTH(1/0)_{i,t}$ interaction in the discretionary accruals model, (4) and a negative and significant coefficient ($p < 0.05$) for the $NRISKS(1/0)_{i,t}$ interaction in the discretionary accruals model. It is difficult to interpret the absolute returns results without a confirmation from the abnormal volume analyses. Moreover, the combined results from the fees and discretionary accruals do not reveal a consistent pattern. The fee results suggest a positive relation between the number of risks and auditor effort, but the discretionary accruals results suggest that clients with a large number of risks (and longer reports) have lower audit quality.

[32] An interesting aspect of the paper by Lennox et al. (2017) is that, for half of their sample, they performed a manual search of risks disclosed by the auditor in year t in the existing company's disclosures in year t-1 (i.e., annual reports, earnings announcements, and conference calls) and concluded that investors were already informed about the majority of risks addressed in the auditor's report. These preliminary analyses suggest that the content of the risks' disclosures is not a significant new source of information.

using a difference-in-differences design with a control group consisting of AIM companies for the investors' reaction analyses, (3) using an alternative difference-in-differences design with a control group consisting of matched United States companies for all of our analyses, (4) examining variation in audit quality, and (4) including cross-sectional analyses examining the incremental effect of materiality disclosures.

Also relevant to this discussion is the work of Reid et al. (2015, 2017). The first study investigates variation in investors' reaction and concludes that abnormal trading volume increased after the implementation of the new rules. The second study examines variation in audit quality and costs. The authors document decreases in absolute abnormal accruals and the propensity to meet or beat analyst forecasts and an increase in earnings response coefficients after the adoption of the new rules. Reid et al. (2017) conclude that the changes in reporting requirements are associated with a significant improvement in audit quality but not with significant incremental costs. We find no compelling evidence that our conclusions are significantly influenced by differences between our study and those by Reid et al. (2015, 2017), except for the results on discretionary accruals, which are sensitive to the calculation of this proxy and sample years (see Section 5.3 below).

5.3 Alternative research design choices for investors' reaction and audit quality analyses

We examined the effect of various research design differences between our study and Reid et al. (2015, 2017), specifically differences in the dependent variables, model specification (control variables), control groups (pre-post tests without control group as well as AIM or US-listed companies as control groups), and length of the sample period (four years versus two years around the adoption date). The relevant results of these analyses are included in the online Appendix C.[33] First, we examined whether our investors' reaction analyses are robust to including a similar set of control variables to those of Reid et al. (2015) and to focusing on pre-post changes for premium companies (adopters) in the two years (t, t-1) around the adoption of the new rules. The results are presented in Table C-1. Using these alternative research design choices, we do not find evidence of an investors' reaction to the expanded auditor's report. Second, following Reid et al. (2017), we also calculate absolute discretionary accruals using the balance sheet method and estimating the discretionary accruals model by two-digit ICB industry groups.[34] Table C-2 indicates that using this alternative proxy we find—consistent with the results of Reid et al. (2017)—a decrease in discretionary accruals only for the first-year of adoption of the new rules in our difference-in-differences model (Eq. 1). However, we do not find a statistically significant result using this proxy after including the two years pre- and post-adoption or adding company fixed effects to the model.

[33] Our discussion is based on the last available updates of these manuscripts, specifically July 29, 2015, for Reid et al. (2015), September 2017 for Reid et al. (2017), and February 2017 for Lennox et al. (2017). Please refer to the online appendix for a more detailed analysis of the differences between our study and Reid et al. (2015, 2017).

[34] We calculate total accruals (TA) using the following equation: TA = (change in current assets – change in cash – change in current liabilities + change in current debt – depreciation) / lag of total assets. Hribar and Collins (2002) show that there are important shortcomings in calculating accruals using this approach. In our main analyses, we follow Kothari et al. (2005) and calculate total accruals as net income – cash flow from operations.

Third, as an inverse proxy for audit quality, we also use the incidence of meeting or beating the analyst forecast consensus, which is defined as an indicator variable equal to one if earnings per share exactly meet or beat by one penny the latest analysts' forecast and equal to zero otherwise.[35] As Table C-3 shows, with this alternative proxy and similar control variables as Reid et al. (2017), we also find a statistically insignificant difference-in-differences effect. Fourth, we also examine whether the earnings response coefficients (ERCs) can serve as a proxy that captures changes in investors' perceptions of the quality of the audited earnings (e.g., Reid et al. 2017; Fung et al. 2017) after the adoption of the new rules. The results of this analysis are included in Table C-4. With this alternative audit-quality proxy and the four-year sample around the adoption date, we also find a statistically insignificant difference-in-differences effect.[36] These combined results confirm the inferences from our main analyses.

5.4 Using a control group consisting of matched US companies

We also use an alternative control group consisting of a matched sample (based on size, ROA, and industry) of US companies for our difference-in-differences analyses of investor reaction (Table C-5), audit quality (Table C-6), and audit cost (Table C-7). The auditor's report in the United States during this period is similar to the old auditor's report in the United Kingdom. Among the advantages of this analysis is that we use larger companies than the AIM companies as a control group; the main disadvantage is that US companies are comparatively farther from the UK economy than AIM companies. In our analyses of investors' reaction, we do not find evidence of an incremental pre-post September 2013 effect for the UK premium companies. These findings support the conclusions that we reached using AIM companies as a control group.

5.5 First and second year of the adoption of the expanded auditor's report

The first year of adoption is interesting because the rule change was somewhat unexpected (i.e., the final FRC rules appeared during that year) and thus constitutes a useful exogenous shock for identification purposes. However, the first year's disclosures might be difficult to interpret for the market or may be useful mainly as benchmark for the second year's disclosures. Including only the first year of adoption and the year prior (t, t-1), we find a statistically insignificant difference-in-differences effect using our four main dependent variables of cumulative absolute abnormal returns, abnormal volume, fees, and absolute discretionary accruals. These analyses reduce the window of our study to the immediate effects of the new rules. Next, excluding the first year of adoption (t), we also find a statistically insignificant difference-in-differences effect using our four main independent variables. Moreover, including year and industry fixed effects in our model (Eq. 1), we find results similar to those documented in our main analyses. Finally, we also examine the potential effects

[35] We note that the inclusion of variables that require analyst forecasts (e.g., meet or beat and unexpected earnings) significantly reduces our sample size. We obtained the latest forecast date from the I/B/E/S detail file and consensus forecasts from Datastream.

[36] In addition, we calculate a pre-post model in which the interaction between $POST_{i,t}$ and unexpected earnings captures the effect of the expanded auditor's report for premium companies after September 2013 and find a statistically insignificant pre-post effect.

of *changes* in the number of risks, focusing on premium companies in the two years after the adoption of the new rules (t, $t+1$).[37] We include an interaction between an indicator variable year t + 1 and the number of risks as well as the main effects in our regression models. The interaction captures the incremental effect of changes in the number of risks. We only find a positive and statistically significant coefficient ($p < 0.05$) for the interaction in the fee model. These last results suggest an underlying association between fees and the number of risks.

5.6 Alternative proxies for investors' reaction

We also use the average bid-ask spread on the dissemination date of the annual report as a proxy for investors' reaction that captures changes in information asymmetry (Lee et al. 1993; Leuz and Verrecchia 2000), computing this value on the filing date scaled by trade price (spread = [ask - bid]/m, where m = [ask + bid]/2). In estimating our difference-in-differences model (Eq. 1), we use the same control variables as the ones in Table 2. In untabulated analyses, we find no evidence that the new auditor's report had a difference-in-differences effect on information asymmetry. Second, we use abnormal return volatility ($AVAR_{i,t}$) in the period surrounding the release of the annual report calculated in the manner of Collins et al. (2009).[38] This proxy captures the effect of information on the volatility of returns using volatility over the estimation window as a benchmark. In untabulated analyses, we find no evidence that the new auditor's report had a difference-in-differences effect on abnormal volatility. Third, we compute abnormal volume, using the market's volume as a benchmark, as the sum of the three-day trading volume scaled by common shares outstanding among all UK listed companies for the period (−1,1) surrounding the filing date of the annual financial statements. Employing this alternative calculation, in untabulated analyses, we find a negative and statistically significant difference-in-differences effect on abnormal volume. Overall, with these alternative proxies, we find little evidence that the new rules affected investors' reaction to auditor's reports.[39]

6 Conclusion

In June 2013, the United Kingdom's FRC issued ISA (UK and Ireland) 700 (Revised October 2012) requiring auditors to include additional client-specific information in their reports for LSE premium companies with fiscal years beginning on or after

[37] For instance, among the 302 companies with available risk data in year t and t + 1, 77 increased and 99 decreased the number of risks in year t + 1.

[38] Abnormal return volatility in the period surrounding the filing of annual report date is calculated using the formula $\frac{\sum_{d\in(-1,+1)} u^2_{it,d}}{3\sigma^2_{it}}$, where $u^2_{it,d} = Ret_{it,d} - \hat{\alpha}_{it} - \hat{\beta}_{it} Mkt_{t,d}$ is the abnormal return on day d, σ^2_{it} is the variance in abnormal returns over the 80-day non-announcement period and $\hat{\alpha}_{it}$ and $\hat{\beta}_{it}$ are estimated over the 80-days non-announcement period (i.e., −45 to −6 with respect to the earnings announcement and from +6 to day +45 with respect to the filing date).

[39] In our main analyses, most of our continuous variables are winsorized at the top and bottom 99 and 1%, the exceptions being the investors' reaction dependent variables (i.e., abnormal returns and volume). This is typical practice in capital market studies (see Leone et al. 2016), but we also find results similar to those found in our main analyses after winsorizing the investors' reaction variables at the top and bottom 99 and 1%.

October 1, 2012. Reports that include this additional information are referred to as the expanded auditor's report. Our research questions address whether the adoption of the expanded auditor's report is associated with a change in the decision usefulness of annual report disclosures, audit fees, and audit quality. Using a difference-in-differences design, we examined LSE premium companies, which were mandated to adopt the expanded report, and a control group consisting of LSE AIM companies, which were not mandated to adopt it. Our analyses include two years before and after the adoption of the new rule. Overall, we find no evidence that the rule changes had a significant effect on investors' reaction, audit fees, and audit quality. Next, we investigated whether variation in the length of the auditor's report, the total number of risks, the number of unique risks (i.e., those not mentioned in the audit committee's report), and the materiality threshold had an incremental effect on the change (before and after adoption of the new rules) in investors' reaction, audit fees, and audit quality. We find little evidence that such variation incrementally influenced the pre-post effect of the new rules. However, premium companies with relatively long reports and large numbers of risks pay comparatively higher fees. This finding is consistent with a positive association between the auditor's discussion of risks and effort, at least to the extent that audit fees capture auditor effort. Nevertheless, the mere *disclosure* of the risks does not cause a change in audit fees or audit quality.

We note some general caveats to our results. First, UK premium companies are relatively large, are at the high end of reporting quality, and have a rich information environment, factors that may decrease the usefulness of the auditor's report. For these companies, the public release of the annual report does not typically provide investors with a significant source of news. For smaller companies, the auditor's disclosures may be incrementally informative. Second, the rule regarding the expanded auditor's report may have a long-term impact not captured by our research design. For example, meaningful long-term changes in the report's disclosures may be informative for investors. Third, it is difficult to isolate the market reaction to the auditor's report from other disclosures in the annual report, including those by the audit committee. Fourth, there may be compliance costs that are not reflected in audit fees. Finally, there could be other direct or indirect effects of the expanded auditor's report that are not captured by our analyses (e.g., changes in investment efficiency and creditor's perceptions of financial reporting risk) and can be the subject of further research.

Our study stands to extend the literature examining the value of audit reports and to inform ongoing regulatory actions. We also hope that our study will motivate the discussion of additional mechanisms for improving the content of the report in the United Kingdom and other jurisdictions. For example, further implementation guidance may encourage auditors to provide comparatively more detailed and engagement-specific information regarding risks and to explain how an audit plan is tailored each year to mitigate those risks. Ultimately, even given the information in the United Kingdom's current expanded report, it remains unclear whether auditors are at the forefront of identifying and addressing key risks. Moreover, there are differences between the FRC, IAASB, and PCAOB standards (e.g., there are differences in the KAMs and CAMs definitions and reporting requirements). Future research may examine the effects of these differences across multiple jurisdictions as well as the consequences of future implementation guidance and ongoing regulatory convergence efforts.

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Appendix 1

Variable definitions

Variable	Definition	Source
$ABRET_{i,t}$	the absolute value of the sum of the three-day abnormal returns in the period surrounding the date on which the annual report is filed, calculated each day as the company returns = (price $close_t$ – price $close_{t-1}$)/price $close_{t-1}$ minus same-day returns for the London Stock Exchange (LSE) value-weighted portfolio	Datastream
$ABRET_EAR_{i,t}$	the absolute value of the sum of the three-day absolute abnormal returns in the period surrounding the earnings release date. Abnormal returns are calculated as the company daily returns minus same-day returns for the LSE value-weighted portfolio	Datastream
$ABVOL_{i,t}$	the natural logarithm of the ratio of the company's mean event-period volume divided by its mean estimation-period volume. Event-period volume is the daily volume in the three-day event window beginning on the day before the report filing date scaled by shares outstanding. Estimation-period volume is measured over the trading period beginning 61 days before the earnings announcement date and ending 40 days later (i.e., 21 days before the earnings announcement date)	Datastream
$ADOPT_{i,t}$	indicator variable equal to one for premium listed companies and zero otherwise	Annual report
$BETA_{i,t}$	slope coefficient obtained by regressing the company daily return on daily returns of the LSE value-weighed portfolio over a 220-day period (−250, −21), relative to the filing date	Datastream
$BIG_{i,t}$	indicator variable equal to one if the company has a Big 4 auditor and zero otherwise	Annual report
$BUSY_{i,t}$	indicator variable equal to one if the fiscal year-end is between December and March and zero otherwise	Annual report /Worldscope
$CFO_{i,t}$	cash flow from operations/total assets	Worldscope
$CHNI_{i,t}$	(net income before extraordinary $items_t$ – net income before extraordinary $items_{t-1}$)/total $assets_{t-1}$	Worldscope
$DACCR_{i,t}$	absolute discretionary accruals estimated using the Jones model including ROA and scaling all variables by average assets	Worldscope
$FEES_{i,t}$	natural logarithm of audit fees	Annual report
$FORSALE_{i,t}$	foreign sales/total assets	Worldscope
$GCO_{i,t}$	indicator variable equal to one if the auditor issued a going concern opinion and zero otherwise	Annual report

(continued)

$INV_{i,t}$	inventory/total assets	Worldscope
$LAG_{i,t}$	days between the earnings release date and the annual report filing date	Regulatory News Service
$LAGACCR_{i,t}$	total current accruals for year t-1(measured as net income before extraordinary items plus depreciation and amortization less cash flow from operation) scaled by total assets	Worldscope
$LENGTH(1/0)_{i,t}$	indicator variable equal to one if the average number of words in the auditor's report in the adoption years t and $t+1$ is above the 50th percentile of the distribution for all companies and zero otherwise	Annual report
$LEV_{i,t}$	long-term debt/total assets	Worldscope
$LOGMKT_{i,t}$	natural logarithm of total market value	Worldscope
$LOSS_{i,t}$	indicator variable equal to one if ROA is negative and zero otherwise	Worldscope
$MATPERC(1/0)_{i,t}$	an indicator variable equal to one if the average materiality in British pounds, scaled by total assets, in years t and $t+1$ is above the 50th percentile of the distribution and zero otherwise	Annual report /Worldscope
$MTB_{i,t}$	equity market value/book value	Worldscope
$NRISKS(1/0)_{i,t}$	indicator variable equal to one if the average number of risks mentioned in the auditor's report in years t and $t+1$ is above the 50th percentile of the distribution and zero otherwise	Annual report
$NRISKSAUD(1/0)_{i,t}$	indicator variable equal to one if the average number of unique risks in the auditor's report (those not mentioned in the audit committee report) in years t and $t+1$ is above the 50th percentile of the distribution and zero otherwise	Annual report
$NSEG_{i,t}$	natural logarithm of the number of business segments	Worldscope
$POST_{i,t}$	period indicator variable equal to one for the two years following the new report requirements (starting in September 2013) and zero otherwise	Annual report
$REC_{i,t}$	receivables/total assets	Worldscope
$ROA_{i,t}$	net income before extraordinary items/total assets	Worldscope
$SALEVOL_{-i,t}$	standard deviation of sales/total assets over year t to year t-6	Worldscope
$SIZE_{i,t}$	natural logarithm of total assets	Worldscope
$USLIST_{i,t}$	indicator variable equal to one if the company is cross-listed in the US and zero otherwise	ADR database of Bank of NY Mellon

Appendix 2

Example of daily returns in the period surrounding the release dates of earnings announcements and annual reports (Ashtead Group plc SEDOL 0053673)

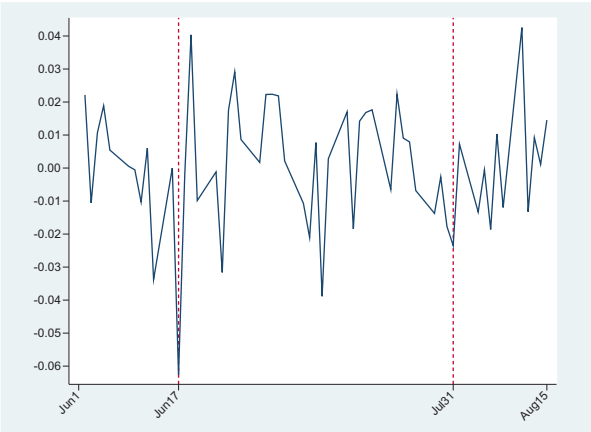

Fig. 9 Daily returns graph including the annual earnings announcement on June 17, 2014 (first vertical dotted line), and the subsequent public release of the annual report on July 31, 2014 (second vertical dotted line). The company's fiscal year-end is April 30, 2014, and this is the first year after the adoption of the new rules

Figures 1 and 2 show the trend in monthly average closing prices and trading volume (in millions), respectively, for the FTSE 100 index from January 2011 to December 2015. The vertical solid line represents the date of the adoption of the new auditor report rules (September 2013). The vertical dashed lines represent the two years before and after the adoption cut-off dates for our study. Figure 1 shows a noticeable increase in the closing price during the period of our study. Figure 2 shows that the first year of our sample (September 2011 to September 2012) was characterized by a period of relatively high trading volume.

Figure 3 shows the trends of daily returns in the FTSE100 and Total AIM indexes (ticker symbols ^FTSE and ^FTAI, respectively). These indexes capture common economic shocks, and their daily returns have a positive correlation of 0.68 (statistically significant, $p < 0.01$).

Figures 4 to 8 illustrate the year-over-year distributions of our main dependent variables, including signed abnormal returns (Fig. 5). The graphs separate the LSE AIM and premium companies by fiscal year. Year t is the first year of adoption of the new rules. Each box and label represent the 25th to 75th percentile range, and the dot inside represents the 50th median. The vertical lines (whiskers) represent the values

Fig. 10 Example of LSE RNS output showing disclosures made by the company, including the earnings announcement on June 17, 2014 ("Final Results"), and the public release of the company's annual report on July 31, 2014 ("Annual Financial Report"), in the first year of adoption of the new auditor's report rules. This appendix shows an example of our dates search. The auditor's report is included in the release of each company's full annual report. The full annual report follows an earlier earnings announcement disclosure. The UK Companies Act mandates that annual reports must be publicly available. Specifically, DTR 6.3 requires a public announcement when the report is submitted to the National Storage Mechanism and available on each company's website. We search announcements of annual filings using the Regulatory News Service (RNS) of the London Stock Exchange as well as the National Storage Mechanism. For the companies in our sample, the earnings announcement is a very complete form of disclosure, containing full financial statements and mentioning whether the auditor's opinion was unqualified. The RNS website can be accessed for searching each company at http://www.londonstockexchange.com/prices-and-markets/markets/prices.htm. An alternative source is the website http://www.investegate.co.uk.

below the 25th and above the 75th percentiles, limited at the 25th and 75th percentile minus and plus 1.5 times the interquartile range, respectively. These figures together demonstrate that, over the years in our study, our main dependent variables remained relatively stable.

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